11

Bringing New Automation to Market

This chapter will lead the reader through the design and development of three new automation products. These machines were the invention of the author over the past 15 years, and they were all created to fill a specific automation need. One is based in the area of precision assembly and fiber optic alignment, while the other two address needs within the packaging community. There was a range of exciting, good, and mixed events along the way, as well as some major frustrations, but there is a wealth of experiences here to be shared to assist automation designers and entrepreneurs.

11.1. CASE STUDY 1: PRECISION AUTOMATION

11.1.1. The Market Need

The precision of an industrial robot has been a research and development challenge ever since they were invented in the 1950s. There was a desire to make robots fast and therefore as lightweight as possible, yet precise for many applications. One only has to look as far as a standard multi-axis milling machine to see a very rigid precision structure that weighs a great amount, so it does not move extremely fast without requiring significant costs. Robot designers were and are concerned for balancing speed, stiffness, arm and payload weight, and precision.

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Robot precision, measured as repeatability and accuracy, has been tested and discussed by industrial robot manufacturers and user groups for many years. Each generation of robots did improve over the previous generation, but many were not sufficient for 0.001 in. or even 1 micron operations. When required, more intelligence was added to the robot and controller, including more sensors for better feedback. Or robot users would add several smaller robot joints and links, almost like a mini robot, to the end of the large robot where the normal robot hand or gripper would be. Attached to the mini robot would be a mini gripper.

The mini robot would use sensor feedback to move or servo to the correct desired position. The sensor could be force feedback, as the large robot and mini robot serial chain were possibly inserting a peg into a hole (the peg and hole analogy is often used as a model of many specific robotic assembly tasks; rarely does one set up a robot system just to place pegs in holes). Or it could be a computer vision system that had some reference marks or cues to align the peg before it was inserted into the hole.

However, this combination of large and mini robot has several limitations:

- Resolution of large robot encoders;
- Deflection of robot arm members;
- Deflection of bearings;
- Gravity effects;
- Servo control system design;
- Friction at joints;
- Dither to overcome friction.

Dither is a common strategy for servo controlled mechanical systems that on the positive side, produces a quicker response. Since the friction at the robot joints is larger at rest (static) than what it is dynamically (kinetic), the frictional effect is minimized by always having the servo motors, and thus the robot arm, moving back and forth on an almost microscopic scale. This can be best observed by feeling the powered up robot vibrate at "rest" with one's hand (touching a robot while its drive system is powered up may be in violation of industrial safety standards). This vibration is the dither effect.

However, the dithering also means that the larger robot holding the mini robot is always shaking. One can see that the mini robot now has a greater problem to solve. The base of the mini robot is moving and twisting in many directions, and the sensor system for inserting the peg has to compensate. If the natural frequency of the larger and mini robots and the control system are not accounted for, the mini gripper might not stand a chance of ever inserting the peg!

So in summary, it is difficult and expensive for a large range of motion (1 or 2 m) and having the robot gripper achieve 1 micron precision. From a machine design view, one is mixing too many orders of magnitude.

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11.1.2. The Closed Loop Assembly Micro Positioner (CLAMP) Device

The first embodiment of the author's invention is called the Closed Loop Assembly Micro Positioner or CLAMP device (Derby, 1989a, 1990). How it was invented is similar to what many inventors will tell you. It was a combination of:

- Seeing a market need;
- Understanding the current solutions;
- Making a few or even hundreds of attempts;
- Divine inspiration.

The author, after using robots to try to solve some assembly challenges for industry, was looking for a better solution for several years. Then one day (while riding the train to the big city) the proverbial light bulb went off. Why not learn from a photographer's tripod? It creates a rigid reference base for taking pictures. And it can be relocated to a new location with relative ease. This is also similar to bracing one's wrist if one has a hand tremor, often found in people past their youth.

In many assembly tasks, the precision work (i.e., the peg into the hole) is not over a large range of motion. The mini robot could be place under the "tripod" and perform its tasks. The tripod would steady the mini robot, and somehow isolate it from the large robot's dither and other challenges. So the large robot could have the limited task of moving the "tripod" from one location to another, and could have significantly less precision.

However, to obtain a repeatable precision base, simply letting a tripod touch a work surface or table top would not be sufficient. Forcing the tripod on the work table surface would be better, but not enough. There needs to be some physical references to align the tripod in a repeatable fashion, so thus the second part of the CLAMP invention, the docking process. To use more commonly found industrial terms, the tripod is called the robot's end effector, and the mini robot a micro positioner.

The CLAMP end effector uses a method of forcing its three legs into three hole features (the process is called "docking") found in the work table surface to achieve a very high precision with respect to the assembly workcell. The most common implementation is to use a conical hole, a conical slot, and a flat plate as the three hole features (using three conical holes would be impractical, since it would mean that the three legs of the CLAMP device must be machined and assembled almost to perfection). The conical hole determines three degrees-of-freedom, the conical slot two degrees-of-freedom, and the flat plat the remaining degree of freedom, thus determining the total six degrees-of-freedom between the end effector and work table surface. This aligning process of two rigid bodies has been referred to as a kinematic mount for many decades.

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The kinematic mount must be precise so as to allow the micro positioner to be as accurate as possible. The ends of the legs can either have a conical surface or preferably a spherical ball. Mating conical elements defines a set of line contact surfaces, and requires very tight machining tolerances. However the set of point contacts of a ball into a conical hole or slot is less demanding, and hardened metal ball components are commercially available (Fig. 11.1). This use of spherical ball ends on the three legs is also useful in that the CLAMP device can be docked into the same set of docking features in three different orientations, 120° from each other. This docking process for a high-precision robotic end effector has been recently duplicated (Würsch et al., 2001).

When the three legs are docked, there is need for some freedom of motion or compliance between the larger low precision robot and the three docking features, otherwise there will be significant wear on the contact points. The CLAMP docking end effector incorporates the use of mechanical springs and several alignment features to assist with the docking and undocking process (Figs 11.2 to 11.5). Conical disks are held into conical holes on the top mounting plate by a series of compression springs. When the large robot pushes down to properly seat or dock the CLAMP device, the springs are compressed and thus allow the lower CLAMP end effector to float freely into the docking features. When the CLAMP device is lifted from the docking features, the springs compress the CLAMP device back into registration to the top mounting plate. This assures that the CLAMP device will be able to successfully dock into the next set of features, and eliminates free vibrations if the coupling was to be simply a set of springs.

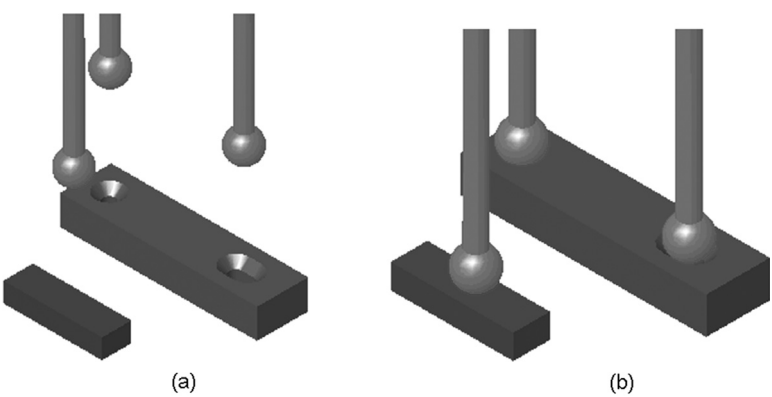


FIG. 11.1 (a) Three legs above docking features; (b) three legs in conical hole, slot, and flat plate

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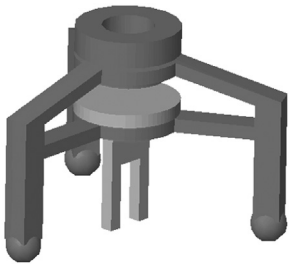


FIG. 11.2 Conceptual CLAMP end effector with two finger micro gripper

CLAMP End Effector Performance

The repeatability of the docking process was measured with a laser interferometer. Measurements were taken while docking with a range of misalignment of the large robot, a range of docking force (and therefore compression of the springs in the top mounting plate), and a range of angular misalignment. The composite results of the CLAMP device are given in Fig. 11.6. These results of less than 1 micron (0.000,04 in.) for repeatability give an excellent foundation for micro positioners and grippers to manipulate.

One limitation of the CLAMP system comes from needing to place a set of docking features around the place where one needs to perform the task. One option is to place the docking features on an XY table's moving surface created



FIG. 11.3 Skeletal CLAMP device docked into docking features

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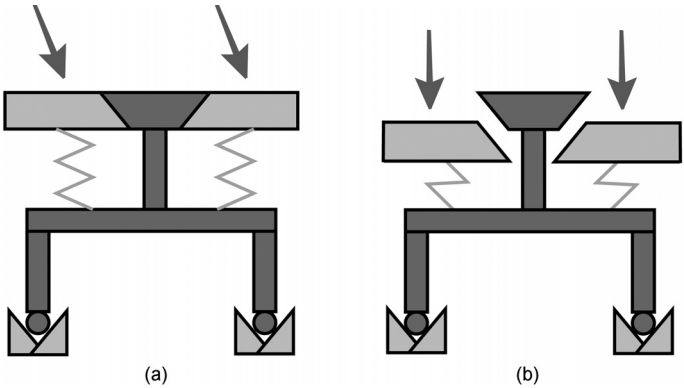


FIG. 11.4 Conceptual top mounting plate compliance device (two-dimensional model)

above the desired working surface (Derby, 1989b). With a hole in the moving surface of the XY table for the micro positioned and gripper to access the working surface, flexibility can be achieved. Another precision implementation would be the coupling of two robot arms. With a CLAMP device on one arm and the docking features on the second arm, a relatively stable precision task can be carried out in free space (Ducoste and Derby, 1990).

11.1.3. CLAMP Production and Market Acceptance

The author was awarded a U.S. Patent (No. 4,919,586, April 1990), and had formed a startup company (GRASP, Inc.). The author remained a faculty member, but hired full-time employees. He hired a talented former graduate student of his, Mr. John McCarthy, as the lead engineer and machinist. John, being the rare

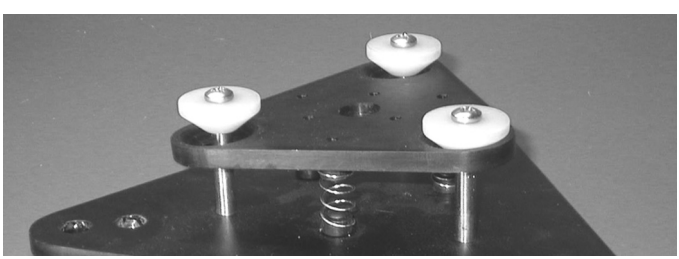


FIG. 11.5 Actual top mounting plate compliance device

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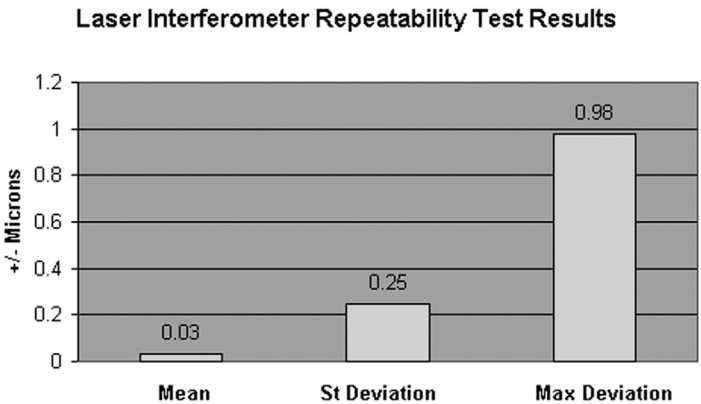


FIG. 11.6 Docking repeatability test summary

combination of Electrical Engineer, Mechanical Engineer, and machinist by education and practical training, was able to design and produce the first CLAMP units.

Since the docking process needed to only hit the conical targets, the larger robot precision was required to only be within 0.25 inch. This was actually a very loose performance specification for robots in the late 1980s, so GRASP also designed and built a moderate speed, lower cost SCARA robot arm. Its major design feature was that the arm members were entirely an exoskeleton frame, allowing for a simplistic manufacturing and assembly procedure. Electric servo motors with harmonic gear boxes were used to drive the arm. It was referred to as the CLAMP robot (Derby, 1992).

The CLAMP end effector was built from aluminum and used off-the-shelf hardened steel balls for the docking legs. It had stepper motor drives for each direction of motion, either translational or rotational, or both, depending on the application. It could be intermittently removed from the larger robot by use of a robotic quick change adapter and a standard gripper temporarily installed for transporting large items.

A PC, several PC interface boards, and some power supplies and amplifiers controlled the CLAMP robot and docking end effector. The PC code was written in the C language, and a user interface written. Mr. Albert Santos, the second full-time employee, handled the control software.

Several CLAMP robots and end effectors were manufactured. They were exhibited at several major assembly robotic trade shows. Interest was good but mixed, since the CLAMP device was a totally new approach to solving precision automation. Much time was spent with each visitor on how the process worked, because no-one had seen this before. The third full-time employee,

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Mr. John Cerveny, was in charge of marketing, and had his hands full trying to educate the potential customer base that as encountered.

However, since sales were slow, and all of the GRASP employees were fresh out of school, a seasoned salesman was hired, Mr. Randy Fields. He knew how the sales cycle worked at our potential customers' factories, and got the list of potential sales to grow to several dozen serious buyers, but timing is everything!

Now it was early 1991, and the Gulf War was under way. The recession of 1991 was kicking into high gear. So despite our sales leader's effort, the company lost one to two customers per week. Not to the competition, but to the fears of recession, and their own potential plant closings. The automation market hit a very soft season, not just for GRASP, but for most of the automation equipment suppliers. Existing major robot companies that had been profitable for years also closed their doors in 1991. There was little that could have been done differently at GRASP that would have meant success. A larger amount of investment dollars would have prolonged the company, but the downturn lasted several years. So the CLAMP device was mothballed for many years, and the author encountered the experiences of company bankruptcy.

11.1.4. The Dockbot

The setback of GRASP did slow down the entrepreneurial effort into automation for the author. But another day of inspiration occurred in the late 1990s, and he was tasked to develop several new inventions. The first of these is the Dockbot (Derby and McFadden, 2002).

The CLAMP end effector is limited in that a larger robot must hold it in place while the smaller micro positioner and gripper does its assembly task. Thus one large robot cannot service additional smaller robots (CLAMPs) simultaneously. It also becomes difficult to coordinate the larger robot to an incrementally moving work surface, such as a pallet conveyor. The novelty of the new docking and locking solution (the Dockbot concept) is the addition of:

- Method of locking the legs of the existing CLAMP device;
- Requirement for the wrist exchange;
- The ability for the larger lower precision robot or automation device to be able to re-aquire the Dockbot, since the wrist interface may have shifted during the compliance motion while docking;
- Device no longer obtains electrical control signals through the larger lower precision robot or automation device via the wrist;
- Now uses connections from worktable for control signal/power; or
- Uses wireless control means (i.e., IR, RF signal);
- Uses stored power;
- Uses onboard microprocessor for control;

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The docking process is very similar to the CLAMP device. The locking process can use:

- Ball and detent methods;
- Spring-loaded rollers similar to many kitchen cabinet latches;
- Permanent magnets and steel plates;
- Electromagnets and steel plates;
- Threaded socket latches — rotary.

Additionally, the preferred method of the locking process does assist in improving the docking process. The spring-loaded rollers option also guides the spherical ball ends of the legs into the docking features, softening the docking process (Figs. 11.7–11.9).

Various Dockbots Solution

Since the larger lower precision robot is not required to be connected to the Dockbot continuously, the lower precision robot may service multiple similar or different devices. This system may be configured as a single SCARA (Fig. 11.10) or six degree-of-freedom (DOF) robot moving multiple Dockbots within its reachable workspace. The Dockbots are not limited to having the docking features parallel

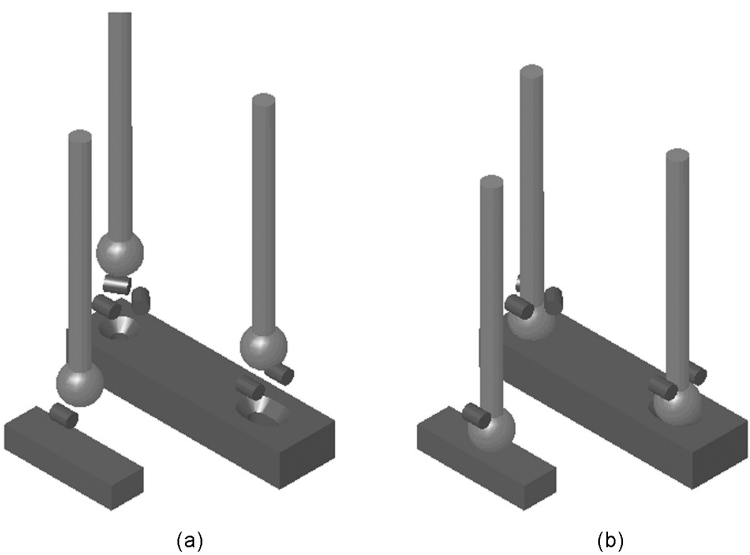


FIG. 11.7 Spring-loaded rollers above docking features: (a) legs above; (b) locked in

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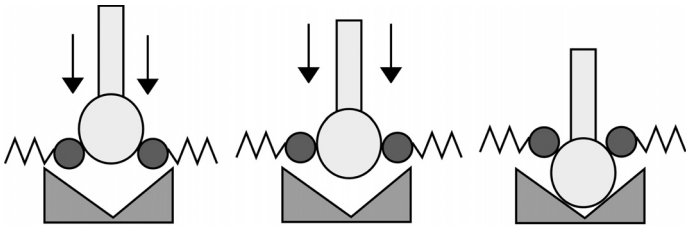


FIG. 11.8 Two-dimensional version of spring-loaded rollers and docking/locking process

to the floor. They could be at various angles, on a vertical surface, or even upside down.

The multiple Dockbots could each have different functions internally, with various micro grippers or operational processes. Each could get its own internal microprocessor program via RF or IR communication, for true distributed robotics. The Dockbots can be effectively parallel processing, where the SCARA or 6 DOF robot is the distribution and allocation device.

The Dockbot's docking features can also be located on multiple pallet surfaces, where the SCARA or 6 DOF robot essentially leapfrogs one Dockbot over a series of other Dockbots to the next available pallet.

11.1.5. Micro Positioners

Micro positioners have jumped from experiments in the laboratory to commercial products. Mechatronic solutions have been very useful in making them a reality.

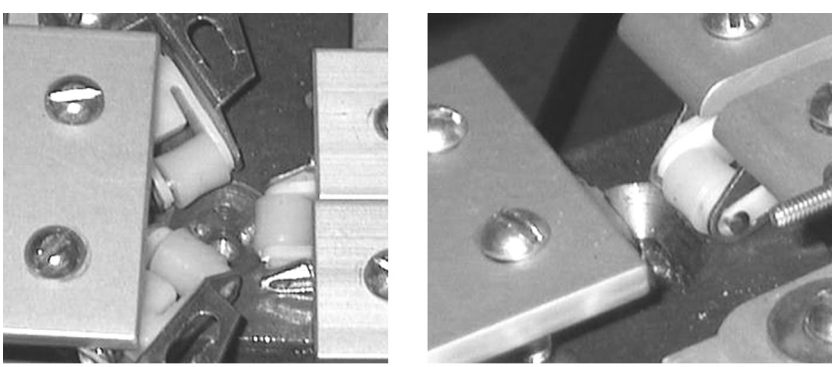


FIG. 11.9 Spring-loaded rollers: (a) above conical hole; (b) above conical slot

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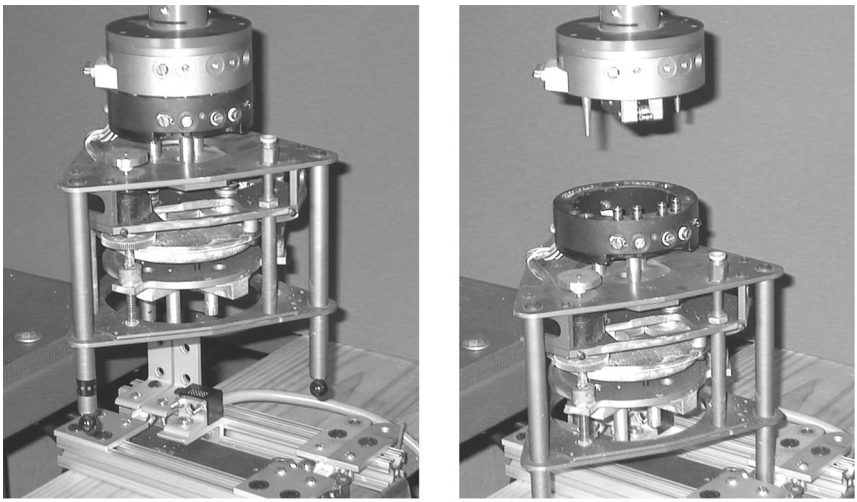


FIG. 11.10 Dockbot being docked by SCARA robot: (a) above (b) in docking features

Many of these devices were developed for aligning fiber optics, which are used for Internet and phone communications. Most have either one or two degrees-of-freedom or direction of motion. The enabling technologies include:

- Micro Stepper motors;
- Voice coil technology (CDROM uses this);
- Piezoelectric actuators.

There are piezoelectric actuators that have one or two axis stages available, with 2.5 cm travel in X and Y directions and 0.1 micron resolution (Del-tron, www.deltron.com/newproduct.htm). There are micro stepper motors available with an accuracy of 0.01 mm per 25 mm of travel and a repeatability of 0.0025 mm (NanoMotion Ltd., www.nanomotion.com). There are also three-axis stage systems (National Aperture, Inc., www.nationalaperture.com) available (Fig. 11.11). A micro gripper would need to be added to all of these for useful work to be performed.

11.1.6. Dockbot Production and Market Acceptance

The effort to bring the Dockbot to market was different from the CLAMP device and the two other machines to be discussed later in this chapter. The goal was to find a technology partner to license the technology, as opposed to full development and marketing within the author's company. There was only limited

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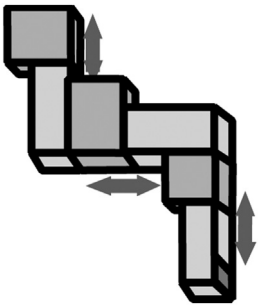


FIG. 11.11 Nanopositioner placed underneath Dockbot: without micro gripper

prototyping performed, and various presentations and conference talks given. The markets explored included:

- Laboratory automation;
- MEMS fabrication;
- Fiber optic alignment.

Those conference attendees handling new technology received the concept of the Dockbot with excitement. Many exhorted the author for the novel approach, and that the industry had suffered from a lack of creativity, the constant cloning of everyone else's machines. But poor timing saw the Dot Com world crumble (fiber optics) and the drug discovery field (laboratory automation) lose money. The MEMS area was slower to take off than was predicted. So the author is still looking for the right partners.

In the future, it might be possible to extend this approach to have the micro positioner of today's Dockbot move an even smaller Dockbot to its docking location to manipulate a nano positioner. This is an area of future research.

11.2. CASE STUDY 2: PALLETIZING

Palletizing is the process of stacking items on a pallet for shipping. The items can be bags, cans, or loose-formed product, but the great majority is cardboard boxes or cases. These cases can have finished product ready to place on the market shelf, such as a dozen cans of beans or 10 boxes of corn flakes. Or they can be filled with assorted products such as letters for the post office.

11.2.1. The USPS Mail Tray Project

It was the palletizing of cardboard trays of letters that first drew the author's attention to this market. In the mid 1990s, the United States Postal Service (USPS)

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was looking to automate the material handling of these trays. Every evening, hundreds of postal processing facilities around the country sort out the first class mail and place letters in reusable cardboard trays bound for various locations (Figure 11.12). The trays are approximately 12 in. × 24 in. × 6 in. in dimension. Filled trays have a cardboard sleeve slid over them, and a plastic band is strapped around the length of the tray to keep the sleeve and tray an integral unit.

The USPS ships most of these trays on several different models of carts (Fig. 11.13). These carts are approximately 6 ft high, and can accommodate three trays stacked side by side per layer. The trays are reused many times, and can become quite ugly looking after only a few uses. The stacks of trays are somewhat unstable, often looking like they could fall over, and the carts' loading was, and in most cases still is, done manually.

The first reaction by the author and many other engineers was to replace the cardboard trays by some more durable and rigid material. France's mail system was automated during the 1990s by changing to a reusable plastic tray with an interlocking lid. This changeover was the logical choice for the USPS also, but being a joint effort of the Government and private industry, the USPS often finds change very difficult. So the automation was to be done with a mix of good and relatively poor trays (in many industrial applications, this would be considered foolish).

The automation is to be configured in an area of the USPS facility that is potentially different in every site. Depending on the location of the site, the normal outside transportation routes for the mail carts, and other factors, the number of mail carts to have trays loaded into them would vary from 8 to 20. The available floor space was also unique in many sites due to the desire to not modify the overall building framework and support columns.

This meant the automation system should preferably be modular (to adapt to every USPS facility) and reconfigurable (to adapt to a new mail route) so that the machine did not have to be scrapped. There was also the internal love at the top administration of the USPS for existing commercially available robots,

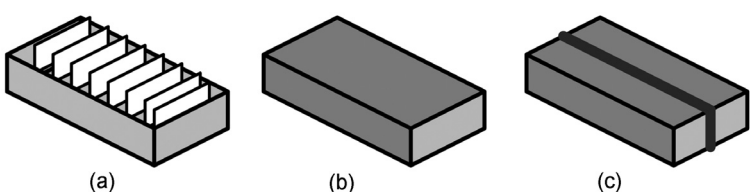


Fig. 11.12 (a) Tray with mail; (b) tray with sleeve; (c) strapped

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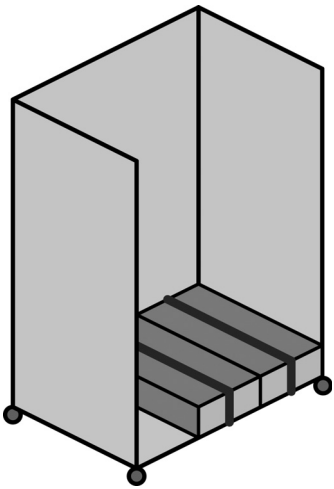


FIG. 11.13 USPS mail cart

similar to what one sees on commercials for automotive companies welding car bodies.

However, this preconceived notion that the solution must take a certain form severely limited the project's success! As will be seen, those commercially designed robots only work best in a certain configuration.

11.2.2. Commercial Robotic Prototype Implementations

The pedestal robot has its maximum reach in a circle (Derby, 1981) (see Chapter 5). To palletize on the maximum number of carts, they too must be in a circle. When the prototype automation system was located in the developmental lab space of a commercial automation house, the circular layout did not seem to be deficient, but when a second unit was actually installed in a USPS site, problems arose. The existing building had support columns that the workcell must accommodate, and the overall traffic flow of people and carts on the floor was a rectangular grid. The need to load and replace carts radially from the circular layout required even more space claims than was originally planned. Simply put, it was trying to fit a round peg into a square hole. The corners of the square were unusable and therefore wasted space. Many of the other future sites had

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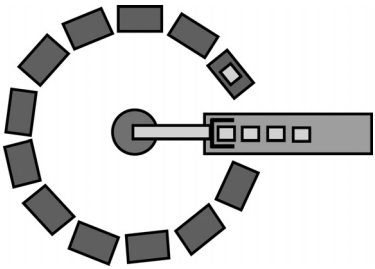


FIG. 11.14 Pedestal robot workcell: top view

support columns in less accommodating locations, so that many other site managers were reluctant to implement the workcell.

Another problem was the number of carts that could be reached by the pedestal robot. A limit of 11 or 12 was all that was possible. However, many potential sites required that more than 12 carts needed to be filled at any one time, so either a second robot system was needed, or manual loading of carts beyond the limit would still be required.

A second prototype robot workcell was configured by the same commercial automation house around an overhead gantry unit. Looking somewhat like an independently supported overhead crane similar to those found in many factory high bays, two column supported rails allowed for travel by a moving horizontal beam. The beam supported a vertical telescoping sliding member that had the gripper attached to its lowermost point.

On first inspection, this robot workcell configuration seemed superior. The supported rails could be as long as needed, so as to load any number of carts. Only more column supports would be needed. However, several problems arose:

- The robot workcell cost became much higher due to the massive robot structure.
- The cycle throughput became an issue, since the gantry supported beam had to travel from the pickup point at one end to the farthest cart in the same cycle time.
- Many of the existing industrial sites did not have the available ceiling space to install the robot.

Both of these prototypes had their champions within the USPS, and during a stretch of many months, a contract of 100 of each automation system was given to two different firms. Neither of these systems was successful due to costs and/or physical constraints. The remaining approximately 7000 possible system locations remain manual to date.

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11.2.3. First Implementation: Linear Track System Design

There were two different answers to this problem invented by the author. The first one was developed when the system throughput requirements were an average of six trays per minute, 10–12 carts, with a 20 lb maximum payload (The Evolution of Geometry and Its Impact on Industry, S. Derby, C. Cooper, Fifth National Applied Mechanisms and Robotics Conference, University of Cincinnati, Cincinnati, OH, paper AMR97-056, October 1997). The fact that the system specifications crept seemingly every few months or so (30 lb payload, followed by 7–10 cycles per minute, followed by 14–16 carts, etc.) all helped to give automation system designers ongoing headaches.

The general automation design strategy used to determine the linear track robot configuration was to:

1. Understand the performance specifications.
2. Look at the various cart/robot configurations.
3. Do analyses based on cycle time and floor space.
4. Create new solutions to the remaining problems.

The most challenging specification was the cycle time.

The various possible configurations of the carts and "robot" were explored. Since the boxes needed to be stacked at all corners of the carts, it was decided that either a linear slider configuration or a SCARA arm design would be needed. The slider was chosen as giving the greatest access to all regions of the cart, and as being the most flexible for redesign. This flexibility proved later to be quite valuable. A second consideration was the option of rotating the linear arm in the horizontal plane, or having a dual arm system. Although higher in costs, the dual arm method was again selected to meet a potentially higher throughput, which was eventually required.

The options of either a linear track or a pedestal base were simultaneously explored. All of the data from the industry, as well as the authors' experience, showed that the linear track system would be the more efficient due to the rectangular world similar to most machining centers. The dual arms were configured to be side by side, in opposite directions, and moved by the same vertical post on the same horizontal track body.

Any potentially higher costs of the robot could be easily balanced by the lesser cost for site preparation and the cost of floor space. The cycle time was somewhat equal, assuming a limit of 12 carts (as per the earlier pedestal robot configuration).

The author's first design was a linear track robot, but with ultimately an improved strategy on the tray pickup locations. The actuators used are similar to those found in industrial robots. The PLC allowed for reprogrammablility, one of the criteria in defining any automation as a robot. This reprogrammablility allows for the workcell to be installed in many of the customer's sites, without the

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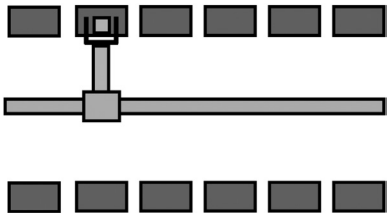


FIG. 11.15 Linear track robot: top view

need for massive plant floor reorganization. Figure 11.15 shows a simplified layout, with the robot loading one of the carts.

The later problem in the design specification process arose from the shortened cycle time. In order to facilitate this requirement, a pair of conveyors was added between the robot and each row of carts, as well as a staging location in front of each cart (Fig. 11.16). The trays were picked up closer to their final cart destination. This made the process the following:

1. The tray barcode would be scanned.
2. The tray would be sent down the proper conveyor, in front of the destination cart.
3. The tray would be pushed by a lever into the staging area in front of the destination cart.
4. The robot would determine an optimum sequence to lift the tray from its staging area and load into the cart.

See Chapter 10 Case Study Number 1 for more details.

This change in plan meant that the vertical reach of the robots had to be doubled. It was easier at this point to increase the length of the slider member than it would have been to add length to a pair of members to a SCARA design. Longer members in a SCARA design would have had interference problems with the cart walls and previously stacked trays.

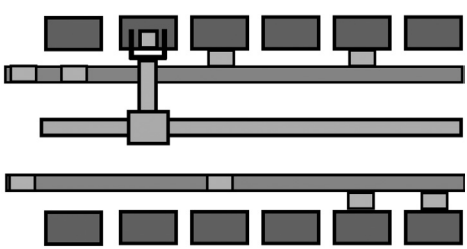


FIG. 11.16 Linear track robot with staging conveyors

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The use of the pair of conveyors and the staging areas increased the throughput so that the desired processing of 14 trays per minute was achievable. Without these additions, it was physically impossible to meet the demand due to excessive speed and practical limitations of material handling.

Prototype Built and Tested

The author worked with American Dixie Group, formerly located in Albany, NY, to design and build the linear track system. Most components were readily available off the shelf. Some combinations, like the 80 ft of hardened steel bearing rails for the main linear track motion, meant that alignment and assembly were not simple. Electric servo motors were used for the three different linear motions. Rotary motion of the servo motors was transformed to linear by use of fiber reinforced rubber timing belts and matching pulleys. Smaller rotary motions that were only 0, 90, or 180° were carried out using pneumatic rotary actuators that have internal programmable stops.

Since the tray and sleeve unit was to be picked by vacuum cups, and this tray/sleeve unit was often somewhat abused from many uses, a powerful vacuum supply was needed. Instead of supplying compressed air to the robot arms and then using vacuum generators, it was determined to have a pair of dedicated vacuum pumps placed on the moving arm base. This was rather unique, since most if not all commercial palletizing robots use vacuum generators due to their lower weight.

Weight was a concern after the linear robot system was built and debugged. The resulting design was a tremendous moving mass running from end to end, and the throughput rate of 14 trays per minute was now demanded to be for more than 12 carts! So the linear acceleration was becoming extremely fast, so much so as to make the technicians worried. So at each end of the larger linear travel direction, a huge spring dashpot/dampener was added to stop any potential high-speed runaway motions.

The unit worked as predicted, but its performance was stretched by the increased specifications. This design became less and less desirable with every new demand. So, the author was not satisfied that the right design was now being used, but what should the new design be?

11.2.4. Second Implementation: The Stackbot Design

Some days an inventor can try as he might, and never come close to anything unique or novel worth patenting. And if one has a new idea, such as what happened when inventing the CLAMP device, one can say it was a very good, exciting day. However, on the inspirational day in the late 1990s when the author

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invented the Dockbot, he also came up with a novel idea for palletizing mail letter trays, called the Stackbot (Derby and Kirchner, 2000).

Assume that most palletizing will be done with similar sized boxes for many hours/shifts without changeover, such as the letter trays into a cart. With this assumption, there are only a limited number of positions in X and Y (parallel to the floor) to which the box must be moved. Why have an expensive set of automation components (servomotors, linear slides, etc.) that allow for unlimited X and Y position access when only a limited number is needed?

The novel concept here is to create a device that uses a selectable multiple cam track plate. The cam track plate's function is solely to determine the X and Y position. It does not carry any of the load of the payload (box or tray), or any vertical load of the device itself. It has a mechanical switch that selects which cam track the payload should travel. The tracks and switch function similar to a railroad train yard, with the various spurs to allow for train car loading or unloading. A dedicated unpowered arm would support the payload weight and give overall rigidity. This arm's sole function is for support, even though it may look similar to a powered robotic arm.

Initial Testing Model

When an inventor conceives of a possibly new automation design, one most likely does not sit down at a CAD system and make production-ready drawings. One must build an initial testing model, or set of models, to prove to one self and any potential investors that the design has merit. Without this proof of principle demonstration, very few people will open their wallets.

Extruded aluminum structures and the various connectors available from websites and catalogs are a great asset. Although the costs are not cheap, one can fabricate many things in one's basement or garage using them with only a few tools. One brand is called 80/20, and it includes rotary and sliding joints that make such concept models straightforward. 80/20 markets itself as the "Professional Erector Set," and the author would have to agree.

An air cylinder was used to power the linear motion. In order to keep things compact, a rodless cylinder was used for the initial testing model. The initial track plate was made from fiberboard, and then a Plexiglas one was cut on a CNC machine when the fiberboard geometry proved to be correct.

There was no capacity in this model to move the track plate or gripper up and down as would be required in practice, but since the concept was to slide either the track plate or gripper using conventional components, it was not seen to be necessary to make this active in the model.

This model, which was almost the correct scale, enough to load three trays side by side in the mail cart, had a trackplate that was quite huge (36 in. $\times$ 36 in.). Perhaps it is not practical if one wanted to access an entire 48 in. $\times$ 48 in. pallet.

This model did its job, both technically to prove to the author and others in the material handling arena, and to excite a company to invest. Morrison Berk-

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shire Inc. (MBI) in North Adams, MA, was looking to expand its product line into new markets. They licensed the design from the author, and with the author's help, designed a commercial unit.

The Commercial Product

Mr. Ned Kirchner, VP of New Product Development of MBI, and Mr. Bill Lyons worked to transform the initial concept to a viable product. Many concepts and variations were sketched and analyzed. Since catering only to the USPS seemed like a hit or miss proposition, the marketing efforts of MBI determined that the Stackbot should also be able to palletize on 48 in. × 48 in. pallets, and if it was to make a good entry into the material handling industry, it should be able to access two pallets at once. (Very few existing commercial palletizing machines work with more than one pallet at a time.) The required trackplate geometry proportions seem unwieldy.

This is where Bill Lyons was clever. When looking at the range of existing mechanisms that have been used over the years (some of them for more than hundreds of years!!), one will find the mechanical linkage called a pantograph (Fig. 11.17). Although mentioned in many Mechanisms textbooks to copy small pictures to be larger or vice versa, it never seems to be used in practice, but this was a powerful combination with the trackplate concept.

The pantograph proportions determine the ratio of input to output. Chosen for a compromise of size and force loading was a 5 to 1 ratio. So the motion of the pin in the trackplate is magnified five times by the vacuum cup gripper, while the trackplate cam tracks see five times the loading of the arm and payload dynamic motion. Figure 11.18 shows a second table top model, with its pantograph.

As can also be see in Fig. 11.17, the forearm link closest to the gripper that transfers the box changes its angle along the cam track path. This means that if the *Z*-axis connected to the gripper is fixed rigidly to the forearm, the box will be rotated by an amount approximately proportional to the arm's travel. This

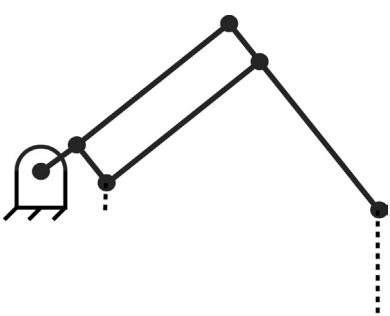


FIG. 11.17 Generic pantograph linkage: cam track path on left, gripper path on right

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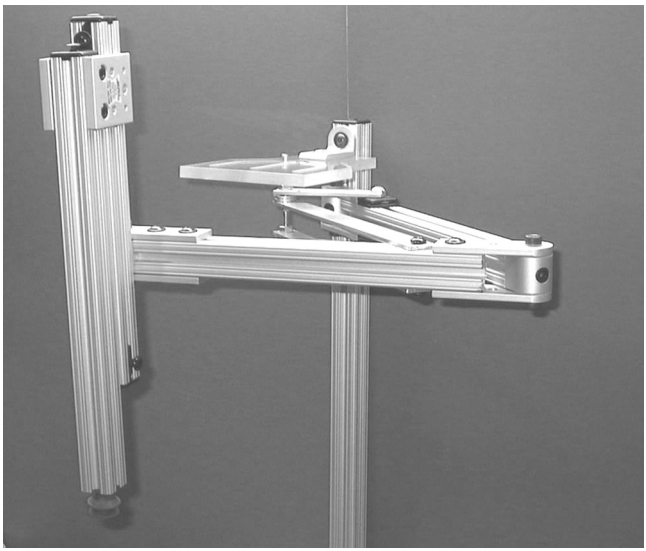


FIG. 11.18 Second Stackbot model

makes things very difficult in placing boxes in a tight, bricklike pattern on the pallet. So the *Z*-axis needs to be able to rotate with respect to the forearm, and it should keep an absolute net rotational relationship of 0° at all times.

The simplest solution, and the one chosen, was to learn from the old-fashioned drafting table attachment that would allow the user to move the "L" shaped drafting guide anywhere while staying parallel to the drafting board edge. This was done in this older implementation usually by cables, pulleys, and sometimes springs to keep the cables taut. A similar setup was done for the Stackbot by using steel belts and pulleys, but the steel belts showed signs of corrosion in a very short time, and actually snapped. Fiber reinforced timing belts and matched sets of pulleys were used and have stood up well (Fig. 11.19).

Then the *XY* position along the cam track can be traversed by applying a linear motion that drives the pin in the cam track of the plate. An electric stepper motor/linear drive unit from Superior Electric was chosen so as to be able to stop at more than one location along the cam track. This is useful when placing boxes along a row on the pallet. A pneumatic rotary actuator was initially chosen as an index device that operated the cam track-switching device. A pneumatic shot pin is energized and locks the track-switching device into the desired directional slot depending on how long the rotary actuator took to move. This was found to be good enough for our factory testing, but in a Beta site testing facility (a bakery) the oils generated from baking were vaporized in the air and coated the Stackbot

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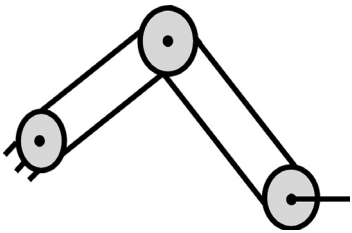


FIG. 11.19 *Z*-axis anti rotation timing belt and pulleys: top view

enough to increase friction and change the performance of the rotary actuator. So, a Geneva indexing mechanism was implemented instead. It was a higher cost solution, but its output was more determined and reliable.

The required vertical *Z*-axis motion for taking a box from its pickup point and stacking it at its place on a pallet can be achieved several ways. The first is to raise and lower the entire cam track plate and unpowered arm assembly from the pickup point and lifting it to the required stacking height. This motion requires a means to have a constrained vertical motion and a method to power it. An electrical motor would be one example, but this has a large moving mass and was rejected.

The second option is to keep the cam track plate and unpowered arm assembly at the maximum height required to stack the top layer. The gripper (suction cups were used) holding the box would have a vertical motion to lift it from the pickup point. An electric motor could be used, but was found to be too expensive. Initially a pair of telescoping pneumatic cylinders with 80 in. of travel was used to lower and lift the gripper, but this proved to be very costly. And due to the fact that the cross-sectional area of the cylinder's internal faces that react to the air pressure are different depending on the direction of travel, the results were not favorable either.

Finally a design for the *Z*-axis that used a standard pneumatic cylinder, and a pair of sliding members and a set of timing belts, created a device that amplified the air cylinder's motion by a factor of two. This gave the results used in production (Fig. 11.20).

The Stackbot can be used to service something beyond the dedicated USPS mail carts if the cam track plate is removable, so as to allow for different sets of palletizing locations. Exchanging the cam track plates is done manually. Track plates are milled using CNC machines. These plates would be held in place by quick release pins, so different boxes or trays could be palletized with a small changeover effort.

Control of the overall device was carried out with a Seimens PLC and a Superior Electric motion controller for the linear stepper motor unit. Incoming

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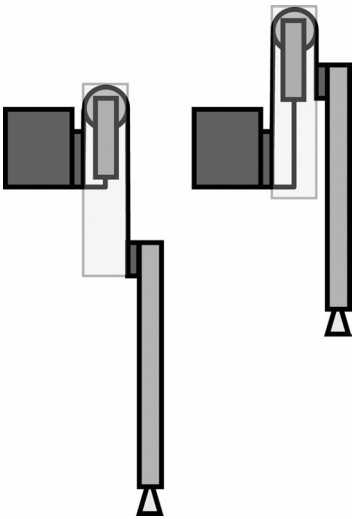


FIG. 11.20 Improved Stackbot *Z*-axis design

box or tray sensor signals would be processed to determine the proper *X*, *Y*, and *Z* height location on the pallet for placement.

In summary, there is less flexibility in the overall Stackbot concept than a standard robotic palletizer, but with the limited number of components and each set of components doing a single task, creates a device that costs less to produce.

What was almost surprising was that a U.S. patent was awarded (Derby, S., and Lyons, B. U.S. Patent no. 6,394,740, May 28, 2002). Not that the author and others had seen the design before, but the invention was a combination of traditional existing mechanisms. There was no enabling technology reason for where the Stackbot concept (Figs 11.21 and 11.22) could not have been invented 75 years ago.

11.2.5. Multiple Stackbots Solution

Another version of this concept would be to create a system of the Stackbots as modules, each servicing its own pair of pallets or USPS mail carts. This creates a parallel material handling system. Many existing palletizing equipment or robotics workcells attempt to cost justify themselves by servicing many pallets or containers at once, but the throughput speeds required to cost justify become unreasonable and unsafe. Stackbots are designed to move more slowly, and be inherently safer.

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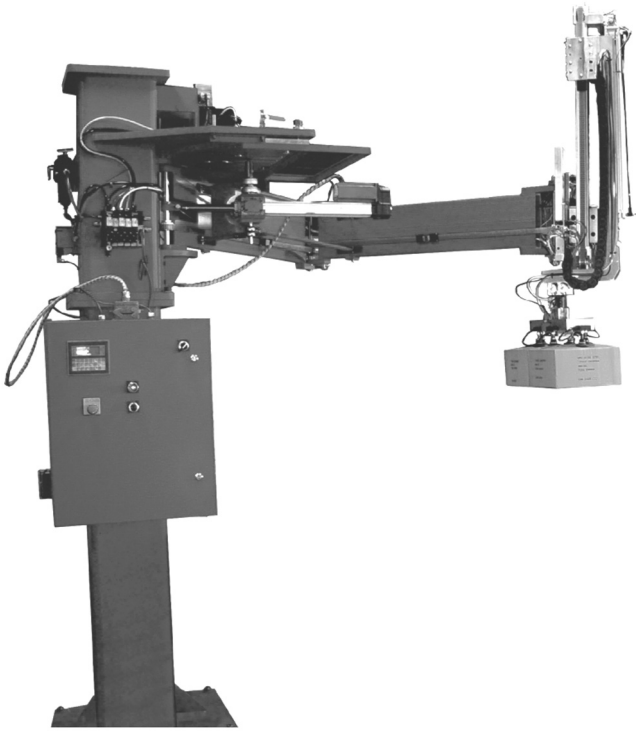


FIG. 11.21 Stackbot

A single or set of multiple conveyors would transport the box or containers to the appropriate Stackbot and pallet location. Each of the individual Stackbots would function in parallel to the other Stackbots (Figs. 11.23, 11.24, and 11.25). The parallel throughput can be much higher than a single piece of equipment or robot, and would work at a slower and safer speed. And being modular in design, additional Stackbots could be added or removed from a system, and configured to handle another task. Controller functions on each Stackbot module would be linked to a central processor, perhaps a PC. This concept can be called Distributed Palletizing (Derby, 2002).

Distributed Palletizing

Palletizing is often carried out throughout the factory by manual means. Many companies palletize at a central location, since their palletizing systems are

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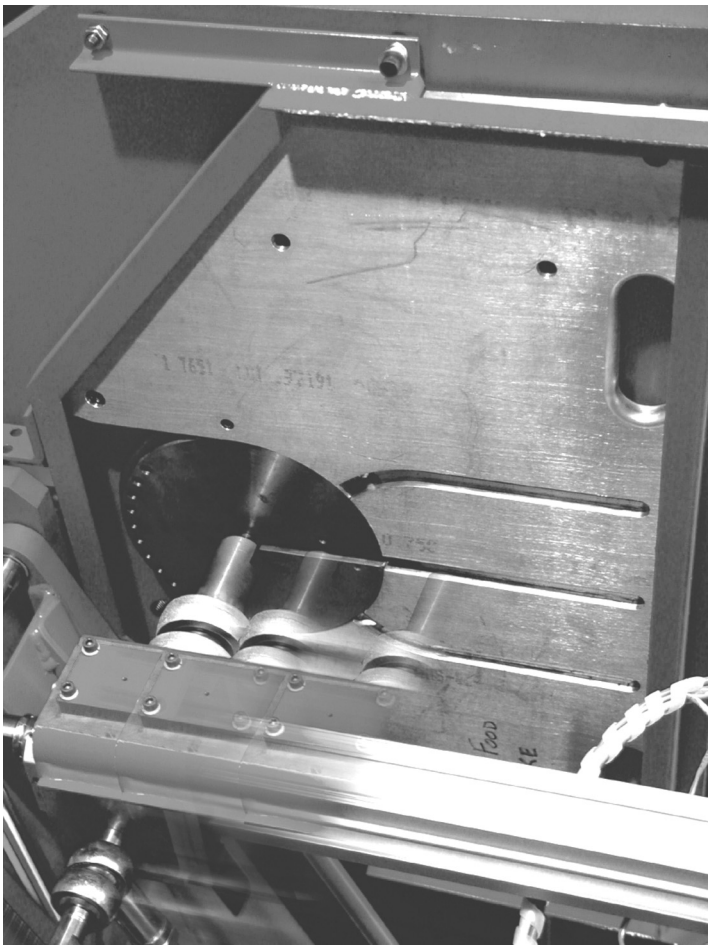


FIG. 11.22 Stackbot cam plate tracks: pin in motion

costly. Some use robotic workcells, and some used dedicated palletizers, which also can be costly.

Chapter 13 will cover many of the Packaging machines including robotic and traditional palletizers, but it is useful to see how the Stackbot, which was conceived and designed to stack letter trays, can be used to revolutionize an existing factory.

A generic sketch of a three-line factory is given in Fig. 11.26. This sketch represents a great number of actual factories visited by the author. A star marks the end of each production line. One could argue that this layout is by definition illogical, but as the author has seen during these site visits, not uncommon. It is

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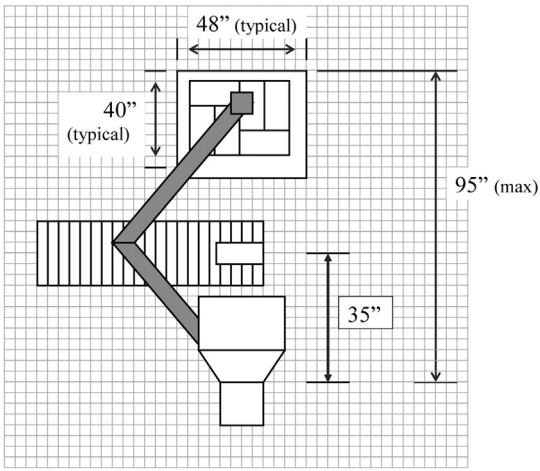


FIG. 11.23 Single Stackbot pallet layout

too costly to reorganize the factory floor and move everything. A production manager has several options for palletizing:

Option 1. One could have a person manually palletize at the end of each line. Depending on the rate of case output and the length of accumulating

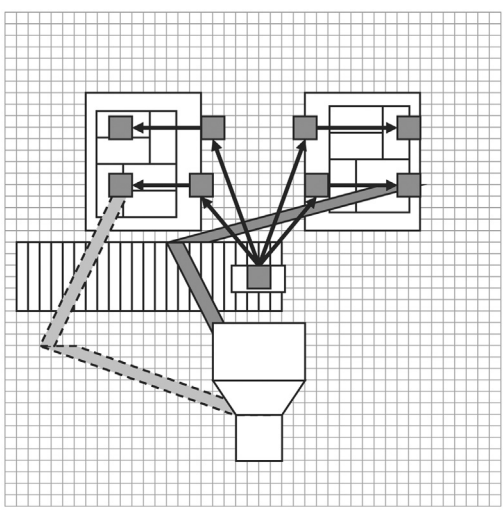


FIG. 11.24 Dual Stackbot pallet layout: with path of box shown

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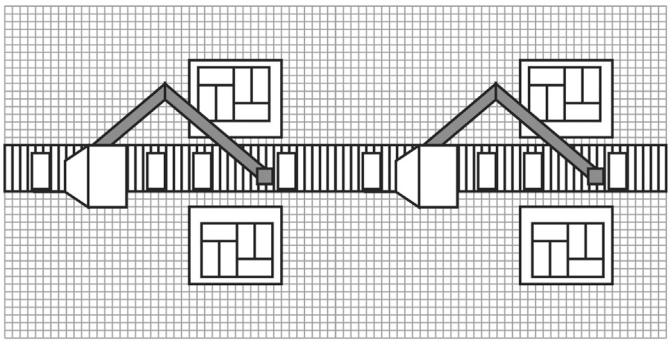


FIG. 11.25 Modular Stackbot layout: Stackbot bases not shown

conveyors, the manpower needed can range from one person per line to one person walking back and forth to cover all three lines. The latter approach is usually unlikely. When a pallet (located at each of the three stars) is filled, a forklift is needed to move it to the shipping area. A path wide enough for pallets is needed (Fig. 11.23).

A suboption here is locating several pallets in the end of the line area, so that the operator does not have to wait for a forklift to change pallets (Fig. 11.27). One is able to keep palletizing on a neighboring pallet. To wait would require an accumulator, and the operator would have to be able to work fast enough to catch up to the production rate.

Option 2. A second option is to install a conveyor system that takes the output of the three lines and either elevates them to a second floor material handling suite, or moves them to a neighboring room. The author has seen both

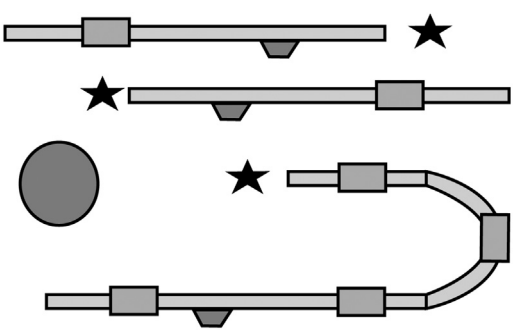


FIG. 11.26 Generic factory floor: stars mark ends of lines

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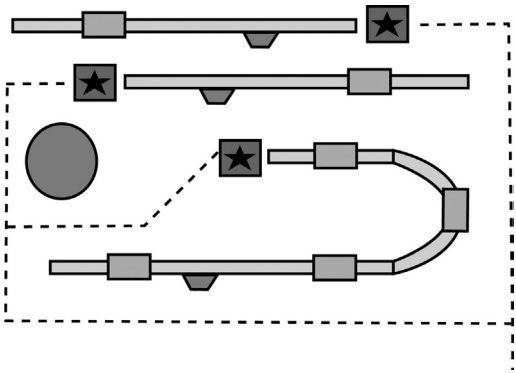


FIG. 11.27 Pallet locations and forklift paths (dashed lines)

approaches used widely. Figure 11.28 represents the conveyor system with the paths marked by dashed lines.

The material handling suite would contain some buffers for accumulation of product, a tracking system if the product becomes mixed as implied in Fig. 11.28, or a set of buffers if each line output is to be kept segregated. A high-speed palletizer would be used to create a pallet load of each product upon receipt of a pallet's worth. A standard robot system would not necessarily have the required throughput.

Option 3. A third option would be represented again by Fig. 11.27, where a palletizer, possible a Stackbot, is placed at the end of each line, around the marked star. This approach becomes practical when the price of the

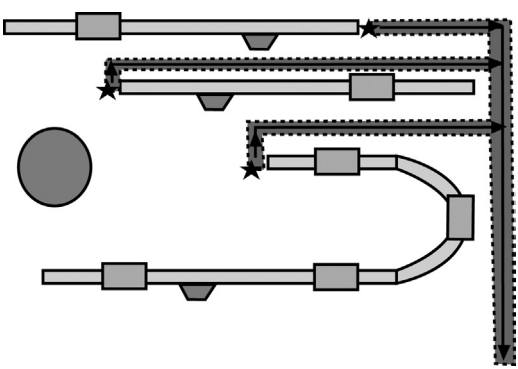


FIG. 11.28 Conveyor system (dashed lines) moves products to a material handling suite located on another floor or in another room

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appropriate capacity palletizer is cost effective. This factor depends a great deal on the rate of product production, and the size of the product.

This product to be palletized could be a small baked good that is first placed in a carton, and then multiple cartons placed into a box. Or it could be a personal health or beauty item, which again is relatively small. In these situations, the resulting palletizing is performed at a reasonably slow rate. These rates may not justify a standard robotic system.

The same suboption as in manual palletizing is then available; that is, using a large accumulator (if needed) for a buffer, so as to switch out filled pallets for empty ones. Alternatively, does one use a pallet conveyor? Or a palletizer that itself can actively handle two pallets?

Use of the third option does define what "Distributed Palletizing" means. Several factories that had implemented a complete material handling suite (Option 2, Fig. 11.28) have removed the complex system due to operational difficulties. Although one plant was not willing to publicly give all of the reasons, it was stated that it "was not worth the operational costs and the headaches." This plant had reverted to Option 1, manual palletizing at the end of each line.

It is reasonable to assume that a centralized system such as Option 2 should be able to operate successfully and cost effectively, so one has to wonder what the other contributing factors were. However, Option 2 does create a complex system that requires a great deal of product tracking and accumulating.

A distributed palletizing system with units handling two pallets at a time allows for the suboption of not having a significant accumulator. This is important when there is little floor space available. Forklifts need to be scheduled only when a pallet is full. The servicing window is much larger and thus production can continue smoothly.

Table 11.1 summarizes this comparison, pointing out the benefits of a distributed palletizing approach in some factories. It should be noted that, in other factories, each line's output could justify a larger palletizing system.

11.2.6. Stackbot Production and Market Acceptance

The development efforts at MBI were on again and off again, depending on what other products they were trying to develop and bring to market. There were some performance clauses in the agreement with the author, but not strong enough to drive the Stackbot development as fast as the author really wanted. Obtaining stronger performance clauses was not simple during the initial agreement negotiations, and neither side envisioned things becoming as slow as they did.

An initial Beta site was set up in a bakery company in Eastern Massachusetts, and perhaps the relationship was started too early in the development

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TABLE 11.1 Palletizing options requirements comparison

	Option 1: Manual	Option 2: Centralized	Option 3: Distributed
Palletizing labor quantity	High	Minimal	Minimal
Forklift labor quantity	High	Moderate	High
Forklift activity location	Distributed	Centralized	Distributed
Accumulator requirements	None if handles multiple pallets	Very large, depends on number of production lines	None if handles multiple pallets
Conveyor Requirements	Limited	Very large	Limited
Floor space requirements	Nominal	Need space for conveyors and accumulators	Nominal
Product tracking requirements	None	Needed to keep correct product on correct pallet	None
Integrated system control	Only signal to get filled pallet removed	Very demanding	Only signal to get filled pallet removed

process. The first Stackbot to go out the door was not tested for any significant stretch of time at MBI facilities. Marketing was pressuring to get a Beta site so as to have a working unit out there to tell potential customers. The Beta site was over two hours away, which limited engineers just driving over to check things out as they should, and the bakery company personnel were not reporting performance issues with the Stackbot on a timely basis.

The Stackbot prototype was also shown at two trade shows to garner customer feedback. Adjustments in minor functions and programming transpired. A sales representative firm was contracted with in conjunction with MBI's dedicated sales force. It was hoped that this rep firm would get into more prospective factories. Things were still progressing slowly. The economy was already slowing down, and very few potential sales were lost to the competition.

Then MBI made some internal corporate decisions to eliminate many of the new products it was bringing to market for another seeming brighter market to hopefully dominate. The Stackbot design was brought back into the author's company. Then a production agreement was established with Cambridge Valley Machining Inc. (CVMI) in Cambridge, NY. (A relationship with CVMI was ongoing from the Trackbot machine, the third case study, to be discussed later.)

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Several automation machine features were determined to be a possible deterrent to future product sales, and a team of CVMI engineers addressed them. These included (problem and resolution):

- Overall arm shaking and perceived lack of stiffness — stiffeners welding onto tubular arm structural members.
- *Z*-axis motion being too violent — improved pneumatic valves and control to better time the air flow amount and direction.
- Troublesome track-switching device — replacing the pneumatic rotary actuator with a Geneva mechanism, as discussed earlier.

The time to perform additional debugging, testing the system to find the weakest links, and in general hardening the automation system to bring it to commercial readiness took much longer than the author and his team had envisioned. These types of commitments are easy to write in a business plan, but harder to actually do, and thus the costs go higher than planned.

But perhaps the biggest challenge to automation builders is the market. The Stackbot was taken to the Pack Expo Las Vegas trade show, held September 10–12, 2001. During the first day, September 10, 100 sales leads (which are a significant number!) were generated. Potential customers were quite excited, and some wanted to buy the unit off of the floor.

Then Tuesday was September 11, 2001, a day that no-one will forget. The 100 leads from the day before might as well have been burned. What was a marginally downward market for automation turned into more than 24 months of being stagnant. Coupled with the economic woes of the USPS, the market for the Stackbot palletizer became much less rosy than what had been envisioned a few years before!

11.3. CASE STUDY 3: POUCH SINGULATION

Singulation is the process of obtaining a single item from a random pile of similar product. It is often found as a bowl feeder presenting one screw after another for automated assembly. Or it can be one of the most challenging automation problems if it is a nonrigid item such as a pouch (Fig. 11.29). As a human assembly worker, it is not too much trouble to grab into a bin and select a pouch and place it in a box (unless the box is moving, as discussed later!), but to replicate this action with standard automation machines is far more difficult.

A machine vision system viewing into the bin or conveyor may see a jumbled mess. Making decisions of where one pouch starts and where the next ends is challenging due to image quality, lighting, reflections, angles of product alignment, and so on (Fig. 11.30). Researchers have been working on this problem for decades, but not cost effectively as far as industry is concerned.

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FIG. 11.29 Pouches of chocolate cocoa powder and snack mix

However, to better understand this problem, it is best to back up a step or two and review parts feeding in general.

11.3.1. Standard Parts Feeding

The most commonly found parts feeder for automated assembly is the vibratory bowl feeder. Depending on the part's geometry, and the likelihood of parts being entangled with each other, some bowl feeders can work fantasticly while others are a thorn in system operator's sides, but a vibratory bowl is not at all flexible for



FIG. 11.30 Pouches as seen with vision camera

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quick part changeover, and is often not satisfactory for the feeder of flexible parts like a pouch.

One could make a limited argument that a standard conveyor is a flexible parts feeder, although it does not determine a part's position and orientation without additional mechanical hardware and/or vision system assistance.

In order to compare the abilities of these standard feeders to more complex feeders, a table is given (Table 11.2). This table distinguishes between required degrees-of-freedom that use servo or stepper motor and control, from degrees-of-freedom that are simple air cylinders with an on/off binary control. Implied is a significant cost difference.

Adept Flex Feeder

One of the commercial flexible parts feeders available uses a combination of conveyors and actuators with a fully functioning SCARA robot (Carlisle, 1997). This design (Fig. 11.31) recirculates parts under a vision camera, so as to use known part orientations when picking up by the robot.

The robot may simply move the picked up part to a registered position on a conveyor or parts pallet, or it may perform the actual assembly process. If it does the former, it becomes a great amount of automation to perform the moderate task. Table 11.2 documents the system for comparison.

Robot and Bowl Feeder

This system uses a vibratory bowl feeder to move parts to a set of pickup points (Fig. 11.32). This mix of older technology and robotics can be used for less total complex automation costs, but is not as flexible. Bowl features would need to be changed to handle other parts (Kohno et al., 1985).

Programmable Reconfigurable Parts Feeder

A different yet functionally similar variation of the Adept Flex Feeder is the Programmable Reconfigurable Parts Feeder as shown in Fig. 11.33 (Gordon, 1994). It uses controlled conveyor belts, a vibratory table and a vision system to singulate parts from each other, and distributes them in random orientation. The robot arm then picks up parts with known orientation. Again, the robot can perform a highly demanding assembly task, or it may simply place the part into a holding template. It is not commercially available as far as the author can determine.

11.3.2. Feeding Pouches

Today many products are packaged in pouches. These pouches are often just one component of a finished product. The cheese sauce for macaroni and cheese, screws in a "some assembly required" kit, and seasonings for instant soup are

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TABLE 11.2 Feeder Types and Degrees of Freedom

Feeder type	Servo DOF	On/Off DOF	Constrained DOF	Degree flexible	Vision	Useful
Bowl Feeder	0	1 Bowl	3 Position 3 Orientation	Very low	None	Yes
Conveyor	0	1 Conveyor	2 Position? 2 Orientation?	High	None	No
Adept Flex Feeder (Carlisle, 1997)	4–6 Robot 2 Conveyors	1 Bucket	3 Position 3 Orientation	Very high	Moderate	Yes
Robot/Bowl Feeder (Kohno et al., 1985)	4–6 Robot	1 Bowl	3 Position 3 Orientation	Moderate	Moderate	Yes
Programmable Reconfigurable Parts Feeder (Gordon, 1994)	4–6 Robot 2 Conveyors	1 Conveyor 1 Vib Table	3 Position 3 Orientation	Very high	Moderate	Yes
Flexible Pouch Feeder (Paster, 1998)	2 Rotational	6 Pistons 1 Conveyor 1 Rotate Bin	3 Position 3 Orientation	Moderate	Limited	Yes

DOF, degree-of-freedom.

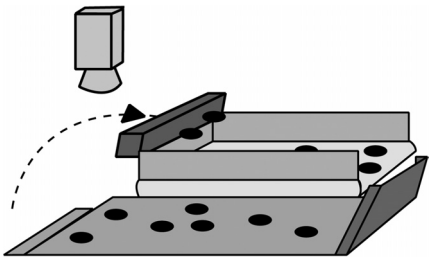


FIG. 11.31 Adept flexible feeder

all examples of pouches packaged together with other components. Often, this pouch is not produced at the location of final packaging. This means that the pouches are shipped, usually in bulk containers, to the packaging site.

Most manufacturing plants today rely on manual labor to place these pouches into their parent container. This is because there is no system for automatically separating and handling them. Classic methods for handling bulk parts, such as vibratory bowl feeders, gravity feeders, and escapement mechanisms, cannot be easily applied to pouches. They are not rigid and deform easily. Methods that work for "hard" parts cannot accept pouches with bent corners or edges, and often cause these defects themselves. Attempts to use bowl feeder technology by industry to singulate pouches have had poor results for these reasons.

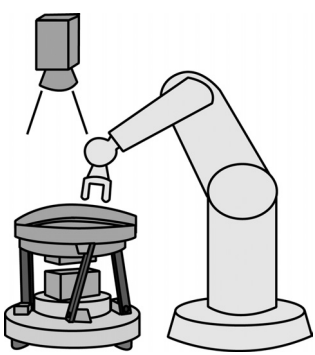


FIG. 11.32 Robot and bowl feeder

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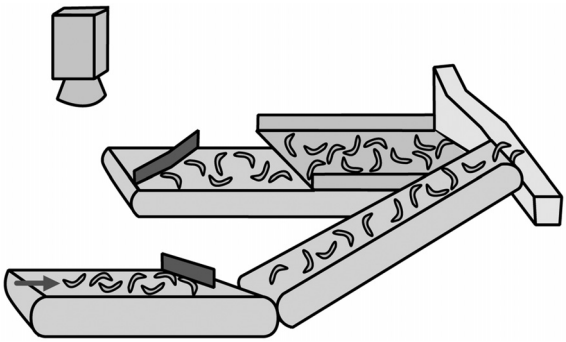


FIG. 11.33 Programmable reconfigurable parts feeder

Feeder Design Strategy: Robot or No Robot

The general strategy of existing commercial feeders, of which almost all would handle flexible pouches satisfactorily, is to manipulate the product, look at it with a machine vision system, and pick it up with a robot. The robot could be either a SCARA or a 6 DOF design. Given, the use of the robot here does allow for more complex operations than a simple loading found in packaging operation, but the requirement to always have a robot may be costly.

These current designs are based on the need to almost always select a product with every robotic motion cycle. If this requirement is removed, then another strategy can be developed. If one examines the operations of the commonly found vibratory bowl feeder, it is based on the "shotgun" effect of letting a larger number of products be cycled by some sorting process, assuming that the output of the bowl feeder will never be starving the next manufacturing operation. This needs to be done while not deforming the pouches.

11.3.3. Vacuum Pickup of Pouches

The use of a vacuum to pick up pouches from a large bin was tested. A vacuum proved an effective means of raising a pouch from the storage bin. The success of this method prompted extensive testing (Norman, 1996). Two suction cups were mounted on a robot to run a series of repeatable tests. The suction cups were lowered into a large bin filled with pouches. Mounting the suction cup with a spring and controlling the distance the cup was lowered controlled the downward force.

The spring was necessary to dampen the impact on the pouches and account for the varying depth of the pouches. As the bin was emptied, the level of pouches

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would drop in the bin. This downward force was important because too low a force would not ensure a good vacuum seal. Too high a force could dent or puncture the pouch.

A series of tests were conducted using two types of pouches, two levels of vacuum pressure, and three types of suction cups. The effect of vibrating the bin to help settle the pouches and prevent dead spots from forming was also examined.

It was difficult to distinguish an optimum suction cup. Larger cups had a higher rate for single and multiple picks, and were more effective at picking heavier pouches due to the increased area. Flexible suction cups were more suited for some pouches than other pouch types. A higher vacuum pressure raised pick rates, but also improved the chances of a multiple pick. A lower pressure was shown to reduce multiple picks, especially for heavy pouches, where the reduced pressure was not capable of supporting multiple pouches. The choice of suction cup was demonstrated to be relatively pouch type dependent. An overall success rate of 50–85% was achieved (Norman, 1996).

The only serious problem in singulating the packets using a suction cup was multiple pickup. When two packets were overlapping and the suction cup hit the edge of the upper packet it was possible to form a good seal across the packets and pick them both up. In some tests, 3–6% of pickup attempts yielded multiple packets (Norman, 1996). This occurrence had to be addressed by the flexible pouch feeder.

11.3.4. Pouch Feeder from a Modified Bowl

The author and his Master's student, Aren Paster, designed an initial prototype pouch feeder that looked like a modified vibratory bowl feeder (Paster, 1998), but it has many additional automation components and functions. It was the first generation automation from the testing performed at Rensselaer Polytechnic Institute.

Flexible Pouch Feeder Design Considerations

The goal of the flexible pouch feeder (Fig 11.34 and 11.35) is to dump a box of packets in, and get a stream of orientated, singulated packets out. Suction cups proved an effective way to manipulate them. The remaining hurdles are to deliver the pouches to suction cup, or vice versa, to orient the pouches properly, and to get them out of the machine. Breaking these requirements down into discrete, individual steps yields the following short list of actions needed to singulate and orient pouches from bulk:

- Spread packets out in large bin;
- Rotate suction cup assembly;
- Lower suction cup assembly;
- Detect 0 vs. 1 vs. 2 packets;
- Detect orientation of packets;

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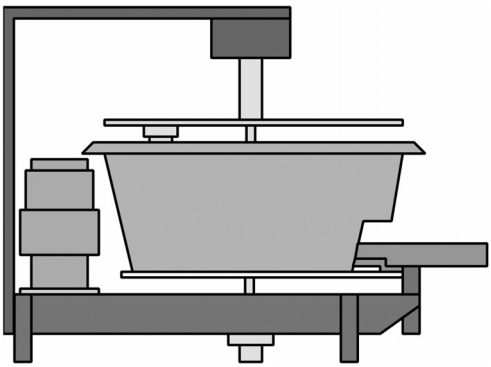


FIG. 11.34 Flexible pouch feeder overview

- Orient packets correctly;
- Feed packets to exit conveyor;
- Operate completely automatically.

Spread of Packets in the Bin

The packets will only be picked up at five discrete repeatable locations around the bin. If nothing is done, all the packets at these five locations will be removed and although many packets remain in the bin, "dead spots" of no packets will develop below the pick stations.

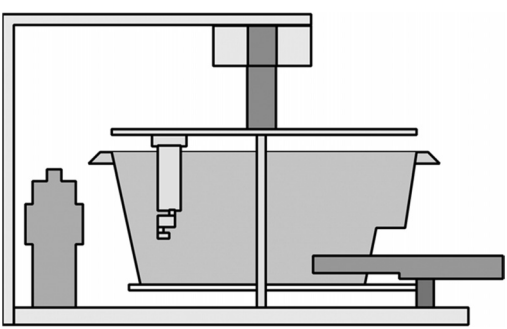


FIG. 11.35 Flexible pouch feeder cutaway

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The floor on the bottom of the bin could be rotated. This would not redistribute the packets greatly, but would move the dead spots away from the pick stations and ensure a more even removal of packets. Implementation of this concept is simple, as the motion need not be tightly controlled. By rotating the floor in the opposite direction of the suction cup assembly, the dead spots will not "follow" under the suction cups. Packets that are added to the system, whether it is from a hopper or those discarded by the vision system, will not accumulate at a constant location in the bin.

Rotate Suction Cup Assembly

The suction cups will be suspended above the bin floor on a rotating disc. This disc will index the cups to six stations equally spaced about the bin. The first five stations will give the suction cup a chance to pick up a packet. At the last station the machine will determine if there is only one packet, if it is oriented correctly, and then send "good" packets out of the system.

A servo motor used to produce the rotary motion would provide the highest level of control for the machine. This system requires a computer, but provides excellent control of the motion. Acceleration and deceleration of the assembly could be precisely controlled, and feedback used to guarantee positional accuracy.

Raise and Lower Suction Cups

The suction cups must be lowered down into the bin to the level of the packets. This will allow the cups to form a seal with the packets, and "grab" them with vacuum pressure. Before the assembly indexes, the suction cup will be raised to reduce the chances of knocking the packet off.

An air cylinder could be used to raise and lower the suction cup. This would require a source of compressed air, either shop air, or a compressor on the machine. Varying the air pressure to the cylinder can control the extension and retraction speed. The force with which the suction cup hits the packets could also be controlled. This force is directly related to the air pressure and thus the speed. Owing to this relationship, to hit the packets lightly, the cylinder must move slowly. This force is important since a high downward force could damage a weak packet. The cylinder could be spring actuated in one direction or air operated in both directions.

If the suction cup successfully picks up a packet at the first pick station, there is no need to try again at the remaining four stations. Lowering the suction cup into the bin may knock the packet off. If the suction cup does not grab a packet at the first station, it should descend at the next station and keep trying

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until it does pick up one. This logical action needs to be controlled, so the machine must sense if the suction cup is holding a packet.

When the suction cup is not covered and vacuum pressure is applied to it, the pressure in the line is slightly below atmospheric pressure. As soon as a packet covers the cup, a seal is formed between the cup and the packet. This lets the vacuum pump suck all the air out of the line, lowering the line pressure significantly. A vacuum sensor could detect this pressure drop, which indicates the presence of a packet. Unfortunately, if valves are used with the suction cups, this may prevent the concept from working. With six suction cups and only one vacuum pump, if one pump was covered, the vacuum force would be transmitted to the five uncovered cups instead of being split evenly. This would provide insufficient force to pick up a packet if too few of the cups were covered. To prevent this from happening, special valves are used just upstream of the suction cups. When the cup is uncovered, a small amount of air is sucked into the cup. As soon as the cup is covered, the valve opens. This exposes the suction cup to full vacuum pressure and it will hold onto whatever began to cover it. This helps to maintain full vacuum pressure in the line and improves the chances of picking up a packet.

Detect Number of Packets

If two packets overlap on the bottom of the bin, and the suction cup hits the packets right on the edge of the upper packet, it is possible to pick up two packets at once. Heavy packets present too much weight for the suction cup, but a lighter packet poses the problem of multiple pickups. This is clearly not desirable, as the goal of the machine is to singulate pouches. This situation must be detected so that multiple packets can be rejected.

When multiple packets are picked up, they rarely overlap more than 30% (Paster, 1998). This means that the area of multiple packets is considerably bigger than a single packet. A vision system could determine the area of the packet, compare this value to a preset size, and easily determine if the suction cup had picked up a single packet or multiple packets. This idea is even more attractive because a vision system would be able to determine the orientation of the packet in addition to the size.

A computer vision system could be used to look at the suction cup and determine the size and shape of the packet(s). This is an expensive option for what needs to be done. A simpler vision system could be built of inexpensive components. An infrared (IR) light source was suspended above a grid of IR detectors. When a packet passes between the grid and the detectors, the detectors are shielded from the IR light. The detectors, which no longer detect light, are in the shadow of the packet. This shadow represents the size and orientation of the suction cups payload.

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Detect and Correct Orientation

Once a single packet has been removed from the bin, it is desirable to have it in a known orientation before exiting the machine. A rectangular packet destined to be placed inside a box may not fit if it is rotated 90°. To provide the packet in a known orientation, the flexible pouch feeder must determine that the current orientation is correctly using sensors, or force the packet into the correctly orientation using "blind" physical manipulation.

Both the computer vision and the simpler vision systems mentioned earlier can sense the orientation of the packet. The straight edges and flat nature of the packets make this a simple task. This information can then be used to rotate the packet to the desired orientation. A second servo motor would be engaged to the suction cup arm at this point, enabling the controlled rotation to be achieved.

Feed Packet to Exit Conveyor

After a single packet in the correct orientation is produced, it must leave the system. It will be placed on a conveyor and carried out of the flexible pouch feeder. How this will be done will depend largely on how the packet is held or located when orientation is complete.

Machine Control

The machine must have some program or code to control the many different actions, inputs, and outputs required to function automatically. Solenoids or air cylinders must be actuated, motors controlled, and vacuum pressure turned on and off. Logical decisions need to be made based on input from sensors and time constraints. A PC was chosen.

The computer control program "sees" what the packet looks like by examining which detectors are in the shadow of the packet. A simple count of how many detectors are covered allows the program to determine the size of the packet. If too many detectors are covered, the suction cup picked up two packets. If only one packet is present, the program checks its orientation. The program compares the length and width of the shadow to the preset packet dimensions. If the packet is crooked or bent, these dimensions will not be correct. The packet is then slowly rotated while the computer rescans the detectors over and over again. When the dimensions of the shadow match those defined in the control code, the packet has been orientated and it stops rotating.

The rotary assembly now indexes the suction cup to a position above the exit conveyor. If the suction cup was holding only one packet and it was able to orient it correctly, it is dropped onto the conveyor. If this did not happen, the suction cup will continue to hold on and drop the packet(s) into the bin immediately before descending at the first station. This allows the machine to give up on a packet if it cannot guarantee that it was aligned and unbent. The

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packet is returned to the bin so that it may be successfully picked up and orientated later. After some time, a collection of bent packets might need to be removed by an operator.

Pickup Efficiency

Two different pouches were used during the experimentation. The first measures 2.5 in. × 4 in. and weighs approximately 0.4 oz. The surface is smooth and consistent and is not prone to wrinkling or puncturing. The pouch is small, lightweight, and solid. The center of mass of the pouch is nearly constant. However, this pouch is very thin, and its small size and flat shape could pose problems during singulation.

The second pouch is produced for a soup product. It measures 4 in. × 5.4 in. and weighs approximately 1.6 oz. It is substantially heavier and larger than the first pouch. The contents are also rather stable, but the center of the pouch is considerable higher than the edge. This bowing around the perimeter provides a difficult spot for the suction cup to establish a good seal. The second pouches have a flexible, wrinkled surface compared to the smoother first pouches.

The suction cup was lowered into the bin with packets beneath it. Tests were run to determine the ability of the suction cup to pick up single pouches (good), multiple pouches (bad), and deformed pouches (good). A number of pouches were deformed to determine the limits of "acceptable" pouch geometry.

The results of these tests were very promising. The concept of handling flexible pouches using vacuum-powered suction cups as demonstrated in previous research (Norman, 1996) has been further proven. These results indicate that the valves used on the suction cups have a significant impact on the success rate.

Usefulness of the Machine

Although the picking of pouches proved promising with the suction cups, the physical layout of the machine became too difficult to pack all of the needed components into the bowl area. The system architecture was just not right. The effort at RPI was placed on the shelf until a better idea would come along.

11.3.5. The Trackbot Invention

Some days are those you remember forever. The same day in the late 1990s when the author invented the Dockbot and the Stackbot, he also invented the Trackbot! There were several other inventions he invented that day that he has yet to develop. It was a very productive day.

The Trackbot design used some of the benefits of the previously described invention, but was more flexible. The earlier concept was always trying to pick pouches at predefined intervals (60° in this case), but the pouches were randomly

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located. After a pouch was picked, it might be placed into a package or box, something that may require the suction cup device to rotate the pouch to get the desired orientation. The desire to pick up at positions defined by sensors or vision systems required a system architecture that was flexible.

Standard Robot Motion

The world of industrial robot arms has been living with the standard robot motion cycle. Defined by some as the "Adept Cycle," a robot and gripper will move a part as shown in Fig. 11.36. The up, over, and place motions, referred to as the "Work Motion," produce a useful operation. The other up, over, and down motion, referred to as the "Return Motion," is wasted time and energy. It is required as the robot and gripper must return to complete the motion cycle. And except in limited cases where two SCARA arms can coordinate their motion, the "Return Motion" also occupies three-dimensional space such that no other robot can move the next part during that time.

This can lead to a very inefficient use of time in many production situations. Or it can lead to the designing of very fast robots (using accelerations and decelerations of up to 10 *g*) that often move or place the carried part so roughly that the part flies out from the gripper, misses its target, or gets damages by the high dynamic (multiple *g*) loading.

The design and performance of a novel robot system called the Trackbot will now be discussed. It was invented by the author, and is patented (Derby, S., and Smith, J. U.S. Patent 6688451, Feb 10, 2004). It has many unique design characteristics, and these characteristics yield a set of motion strategies unlike most single robots today (Tsai, 1999; Groover et al., 1986; Craig, 1986).

The general application area is to be able to pick up a product part from a collection of parts along a conveyor. There are many other variations to this description, and some of them will be addressed later.

Trackbot Design

When one hears the word "track" associated with a robot, it is usually a conventional robot placed on a linear track to extend the effective reach. Some tracks are

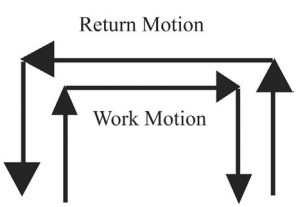


FIG. 11.36 Standard robot trajectories for most current robot configurations

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based on the floor, some overhead (Kato, 1988, 1992; Jerue and Walsh, 2000; Vogt et al., 2001). The system described here is nothing like these standard configurations.

The Trackbot robot system (Derby et al., 2002) has a rigid track to carry the multiple individual robot heads, or Bots. A constantly moving drive chain is used as the motion source so as to not have a drive motor on each Bot, thus saving weight. Figure 11.37 shows the overall concept.

The product parts are represented as simple discs (Fig. 11.38), but in practice can be any size and shape part, potentially a mixed group. A mixed group might be handled best by having dedicated Bots for each type of part, that is, one Bot has a suction cup while another has a gripper to pick up long thin parts. The overhead vision camera represents the possibly use of multiple cameras in order to have a field of vision large enough to recognize the location of the parts on the conveyor.

The moving drive chain runs at a constant speed, while each Bot has the mechanisms to "grab" or "release" from the chain, conceptually similar to a ski lift.

Although perhaps obvious, this design does not permit one Bot to pass the one ahead of it, and direct contact of Bots would constitute a collision. The overall system computer needs to calculate where each Bot will be over time, so as to avoid these collisions. So the actual embodiment of this concept has many concerns.

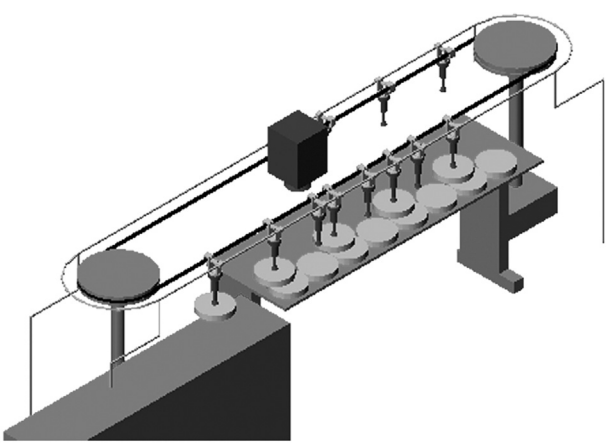


FIG. 11.37 Overall Trackbot system layout. Eleven Bots are on the oval track. Product parts being moved are shown as discs. Many discs are located on the input conveyor. The output conveyor is presently empty. A video camera is located at the midpoint between the two chain drive wheels.

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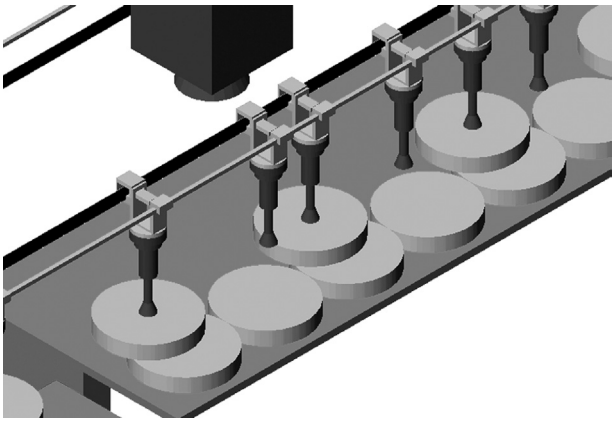


FIG. 11.38 Conceptualization of the Bots being held on the lowermost track and being propelled by the neighboring (thicker) drive chain. The disc products would most likely be more jumbled in a true application.

11.3.6. Initial Test Model

Similar to the early development of the Stackbot, an initial test model of the Trackbot was developed (Fig. 11.39). It was a benchtop unit that had a single straight section of track, and a timing belt on pulleys, much smaller in diameter than the proportions shown in Fig. 11.37. The simple Bot has an onboard Basic Stamp microprocessor, and it was programmed to wait, grab the chain for a few seconds, then release. It literally pinched the timing belt using rubber contacts to obtain motion. The Bot's motion was limited to 2 ft of travel, and had to be manually reset to the starting area after each run. What it demonstrated was that pinching a chain or belt would be unsatisfactory for any length of service. The pinch points showed excessive wear with limited use.

Prototype Production Trackbot System

Shown in Fig. 11.40 is the prototype Trackbot robotic material handing system. Shown are four Bots on one of the straight sections of an oval track. The overall system is 13 ft long, 5 ft wide, and 6 ft high.

Each Bot has a telescoping *Z*-axis of motion with a vacuum cup at the bottom. Each Bot has its own internal Pentium-based single board computer. An IR communication device is located near the top of each Bot, allowing wireless communication. Upon the passing of the IR station, the programmed instructions for each Bot are updated. After receiving such an update, each Bot operates autonomously.

The rail and chain system is shown in Fig. 11.41. It consists of two metallic conductive rails and two chains. The conductive rails allow for 12 V power to be

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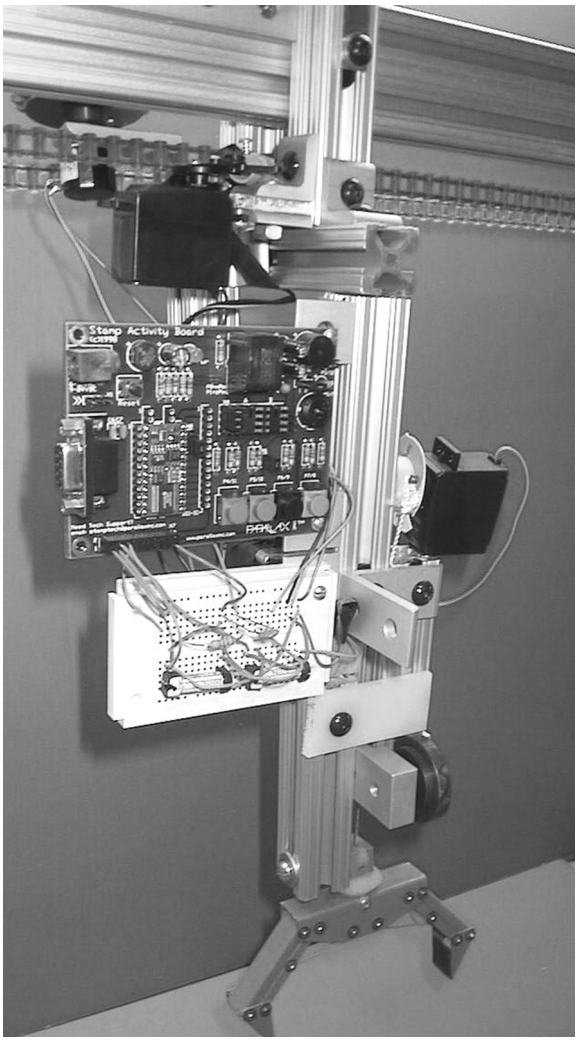


FIG. 11.39 Initial testing Trackbot model

supplied to each Bot through their conductive wheels. The upper chain remains stationary, while the lower chain moves at a constant speed. Each Bot has two cogs that are always engaged into the chains, and are actuated by a clutch pack. The lower cog is clutched to restrict its rotation to enable movement along with the chain. The upper cog is rotating while engaging the upper chain, giving positional information. Releasing the lower cog's clutch and

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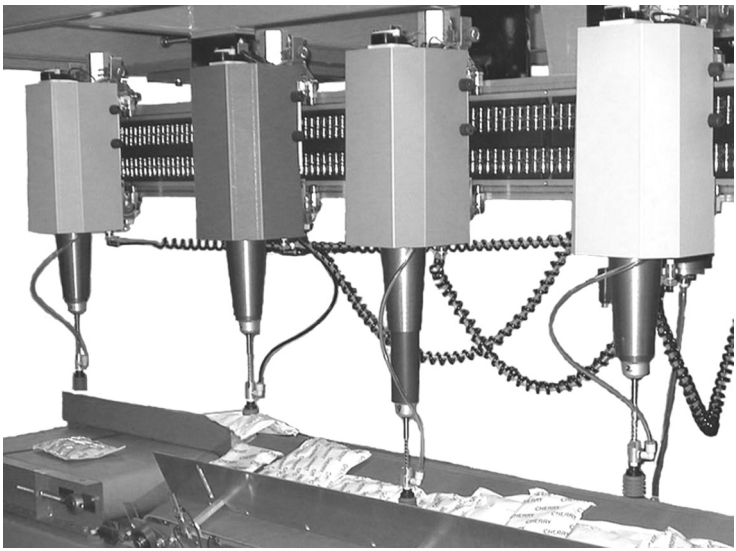


FIG. 11.40 Actual Trackbot material handling system. This application is using suction cups to singulate filled pouches. Multiple pouches are randomly located on the input conveyor, while two singulated pouches have been placed on the output conveyor at the left.



FIG. 11.41 Original Trackbot rail and chain drive system. The upper and lower rails constrain the Bots. The lower chain moves and is used to propel the Bots. The upper stationary chain is used for braking and linear path position feedback.

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clutching the upper cog achieves stopping. This applies a braking action in a repeatable way. Engaging the lower cog produces the acceleration and eventual matching of the drive chain. The overall robotic repeatability is designed to be ± 0.25 in. since the prescribed tasks do not require any significant precision. Any backlash due to the use of a cog and chain is insignificant.

11.3.7. Design Considerations

The reasoning behind this design was the following:

1. It was assumed that vacuum cups on the moving Bots would still pick up pouches. This meant that one either had to:
   (a) supply compressed air, for a vacuum generator;
   (b) supply vacuum directly; or
   (c) generate a vacuum on the Bot by a locally placed electric powered motor.

   Choices (a) and (b) require hoses that could connect the Bots to a central hub, while choice (c) added more weight and space claims. No off-the-shelf small lightweight vacuum generator could be found, so that eliminated choice (c). Having extensive tubing handle a vacuum (choice (b)) is not usually recommended, since the hosing can get sucked flat if not properly designed and installed, and it takes longer to achieve a vacuum, since the entire volume of the hose needs to get to the 8–10 psi pressure drop. Choice (a) was the best of the group. It still required a custom rotary hub to be designed.
2. The Bots needed to have electrical power supplied to them. The most effective amount would be 12 V DC. The three choices were:
   (a) have an electrical cord tethered to each Bot;
   (b) use conductive rails similar to many commuter trains;
   (c) generate electrical power from the motion of the Bots.

   Choice (a) requires an electrical slip ring, which has a history of contact wear, electrically noisy connections, and maintenance woes. So this choice was left as a backup. Choice (c) would be only intermittently generating power while the chain drive was propelling the Bots. It would be difficult to implement, but it could work. Again weight and space claims ruled it out. So choice (b) required that the rails be electrically isolated from each other, and supplied with a 1000 W 12 V power supply for the 12 Bots. The 1000 W amount was a worst case loading determined during the Bot design. There were still many questions at that time, and it was later found to be excessive.
3. The Bots then needed to pick up electricity from the rails. The original design (Fig. 11.41) included round cross sectional rails. This would

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allow for the wheels (Fig. 11.42) to stay in contact whether the Bot was traveling on a straight piece of track or a curved portion. The Bots had a suspension system to hopefully assist with the dynamics. Contact on both sections of track is not a trivial matter, since the wheel carriages must float while keeping the cogs of the clutch pack engaged with the drive and breaking chain. The initial idea chosen to implement were wheels made of electrically conductive silicon rubber. Although they worked surprising well electrically, the Bots basically "blew out the tires" and shredded rubber lay under the track after only a few laps. Then the wheels were made from copper, which did not exhibit the best wear characteristics. Copper filings filled the track too often.

4. The Trackbot prototype was designed to be both an evaluation and testing machine, and a machine to bring to a trade show. A series of internal conveyors was integrated into the machine, solely to recirculate the pouches after picking. No one wanted to have to baby sit the machine for five days at the trade show by moving pouches from one end to the other every three minutes.
5. The vacuum cups would work well for lifting chocolate cocoa powder pouches, but after many reuses in testing and at trade shows, the pouches would break down. Chocolate powder was found on all of the recirculating conveyors, and did get sucked into the vacuum generators. It was desired to grab heavier pouches with a slippery metallic cover, so a switch was made to a more powerful and better filtered vacuum generator from Piab (Hingham, MA, USA).
6. The chain drive system required a pair of chains, one for propulsion, and one for braking. But both chains required the same pair of 3 ft diameter cogs that form the loop. Concern was raised that the propul-

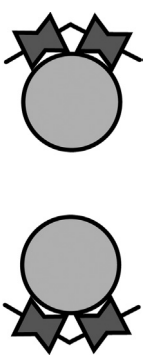


FIG. 11.42 Original wheel profile

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sion chain would receive more wear, and would be likely to stretch. So since the entire design of the clutch pack was reversible in its operations, the Trackbot was designed to be able to swap the roles of the chains. After so many days of operation, a set of toggles would be switched to change which chain and pair of cogs would be the driver and which for the braking. The only real additional component costs were that the Bots would need to have rotary encoders on both halves of the clutch pack, and that the Bot's software would have to intelligently determine which chain was the motion source.

7. The initial concept demonstrated at the Pack Expo trade show in Chicago, November 2000, had a set of line sensors to determine if pouches were present. As seen in Fig. 11.43, this is similar to a multiple implementation of how most grocery market conveyor belts operate. This was found to be insufficient, since for pouches of any real size or weight, the center of the pouch needed to be obtained, and the sensors only indicated a pouch's presence.
8. The Bot's *Z*-axis rotation was initially seen to be not important to implement. After the successful picking of a pouch, it could be placed on an output conveyor and rotated thereafter, but too many potential applications suggested that this operation was not the desired one. So the *Z*-axis rotation was added to the production design.

Redesign Directions

The round rails, wheels, and Bot suspension proved to be unsatisfactory. The wheels did not stay in contact 100% of the time. The transition from straight

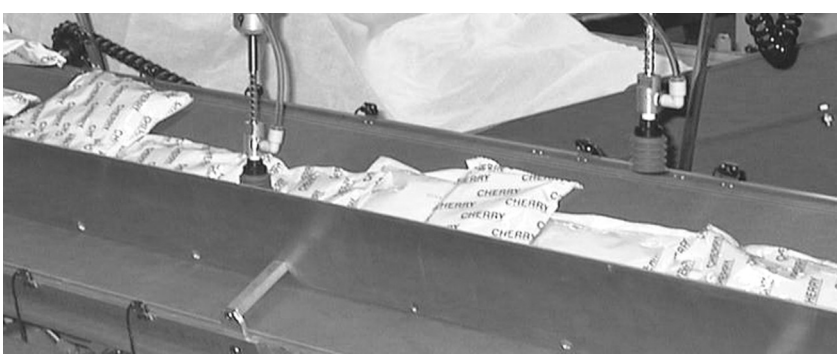


FIG. 11.43 Multiple sensor operation on the conveyor: note holes in metal plate

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motion to curved motion at 4 ft per second causes many an electrical supply gap. Even by added large capacitors to the Bots, the power would drop enough that the onboard CPU would drop and reboot, a definite problem!

So the rails were made to be of rectangular cross-section, and the wheels replaced by commercial cam followers (Fig. 11.44). These followers also conducted electricity.

So unlike a servo motor drive unit found in most robotic actuators, this dual clutching drive system (Figs. 11.45 and 11.46) gives a repeatable but non-variable response. It does also provide a much more cost-effective solution. However, because of this dual chain drive method, some unique opportunities arise for motion analysis.

11.3.8. Trackbot Motion Strategies

To better understand some of the desired motion strategies, a brief look at some of the general uses is needed. In Figs. 11.47–11.49, the square blocks represent the Bot locations. The solid rectangle represents the infeed conveyor and Bot pickup locations. The striped rectangle represents the output conveyor. Figure 11.47 shows the use of one side of the Trackbot, while Fig. 11.48 shows the using of both sides. Figure 11.49 shows six infeed conveyors, each containing a different product to be combined at the striped output conveyor. This process is often referred to as "kitting."



FIG. 11.44 Revised Bot wheel profile

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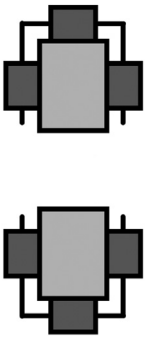


FIG. 11.45 Revised track rail system

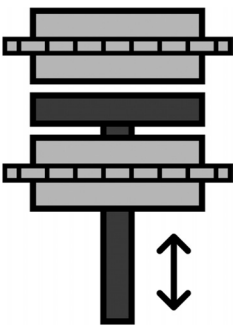


FIG. 11.46 Chain, cog, and clutch pack design

Motion Assumptions

The motion of a single Bot will now be discussed. For most applications of the Trackbot the product pickup point requires that the Bot come to rest. A generic velocity vs. time plot is shown in Fig. 11.50. Although the configuration of the two

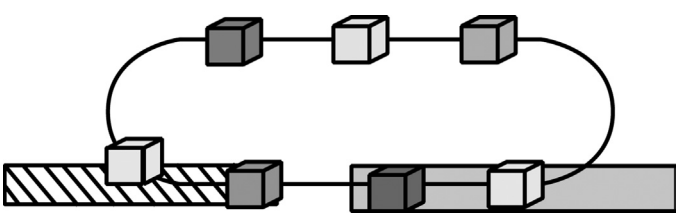


FIG. 11.47 Conceptual representation of seven Bots, rail, and chain drive, input and output conveyors. Bots are moving clockwise

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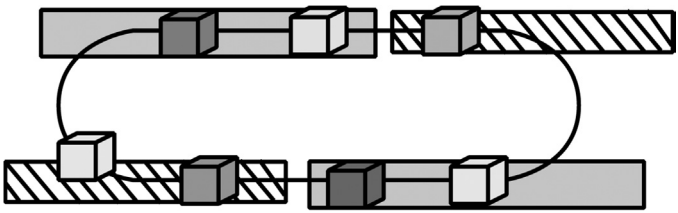


FIG. 11.48 The addition of a second input and output conveyor

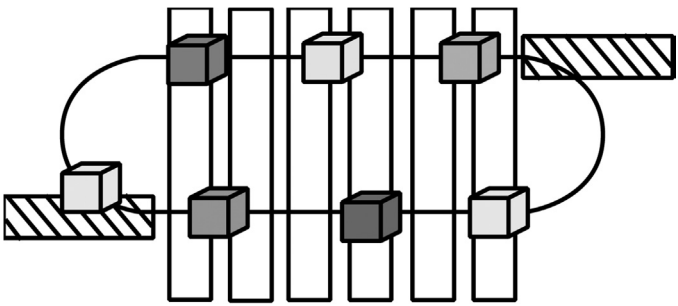


FIG. 11.49 The addition of a second output conveyor and the use of six input conveyors used to create product kits

chains is one constantly moving and the other fixed, the engaging of the cogs' clutch packs (Fig. 11.46) produces a noninstantaneous effect. This is fortunate since the desired Bot motion is smooth rather than screeching to a halt or to full speed.

It should be noted that for most of the applications in material handling concerned in this paper, the product can be placed while the Bot remains in motion. So for one productive looping of the track, the Bot would come to a stop only once. When a second Bot has been added to the same track, the timing of this one stopping location becomes the challenge.

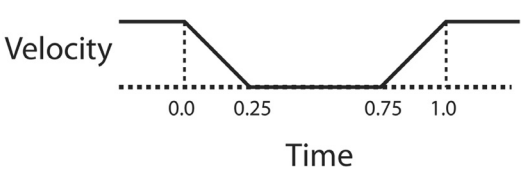


FIG. 11.50 Velocity profile along track vs. time. Zero velocity segment corresponds to *Z*-axis travel time to grab product, 0.5 sec. Maximum velocity is 4 ft/sec.

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FIG. 11.51 Path of two Bots on the same track. Second Bot does not enter deceleration, activity, and acceleration zone until first Bot is back up to speed

It is this strategy of two Bots where the improvement over standard robots is seen. Where the time and space of a standard robotic motion as shown in Fig. 11.36 has the wasted return motion, during that time a second Bot can be performing its task and increasing the overall system throughput.

Part of the overall strategy is how conservative a Trackbot system programmer wants to be. If a Bot is going to stop at the location represented by the star in Fig. 11.51, and the entire zone initiated by the signaling of the lead Bot to begin braking, and concluded by the Bot again achieving the overall chain speed, is to be kept as a "No-Bot" zone, then the rule base is graphically given by Fig. 11.51. If, however, the system programmer is less conservative, the next Bot can get closer, more representing bumper-to-bumper traffic on a congested highway as seen in Fig. 11.52.

After some review of these constraints, a preferred strategy to handle a multiple number of Bots is to operate them in a wave. The four Bots in Fig. 11.53 are programmed as a wave. The absolute distance between each consecutive pair of Bots may be adjusted within some higher level of programming. Bot-to-Bot

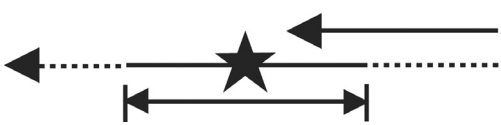


FIG. 11.52 Path of two Bots on the same track. Second Bot does enter deceleration, activity, and acceleration zone before first Bot is back up to speed. No factor of safety

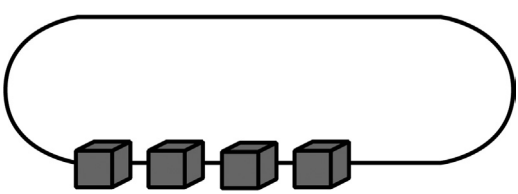


FIG. 11.53 Wave of four Bots on the same track. These Bots are coordinated to keep together to maximize throughput

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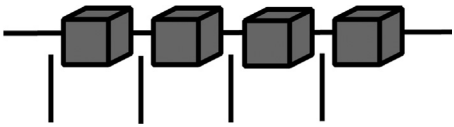


FIG. 11.54 Zones of one wave of four Bots. Each Bot is programmed to go a targeted position within its zone

collisions must be avoided by the overall system computer, but the average over time is a constant.

It will be shown that the maximum number of Bots and waves will be dependent on track length, chain speed, and stop time.

The actual Trackbot Bots are 6 in. wide. A nominal Bot-to-Bot centerline distance is kept at 12 in. Small variations of ± 2 in. within each Bot's zone (Fig. 11.54) can be programmed to pick up randomly located parts on the input conveyor.

11.3.9. Motion Analysis

But how does this compare to a traditional robot and when is the Trackbot a better option? To address these questions a comparison of a robot to a track full of Bots will now be presented.

Single Bot

For comparison purposes, the discussion will continue to base its assumptions around the existing prototype Trackbot. One of the current mechanical design limitations is a minimal radius of curvature of 10 in. (the actual prototype has an 18 in. radius).

Another limitation for comparison in this paper is the assumption that the Bots do stop to pick up a part, but deposit it at the current maximum speed of 4 ft/sec. Given that:

- 4 ft/sec speed chain;
- No-Bot zone of 1.5 ft;
- No-Bot segment takes 1 sec.

A circular track of 5.5 ft circumference makes sense. The 5.5 ft includes the 1.5 ft of the No-Bot zone. During the first second the Bot travels 4 ft, the following second the pickup sequence. The part is to be placed on the opposite side of the circle from the pickup point. The results are shown in Table 11.3.

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TABLE 11.3 Throughput Comparison

Bots vs. robot	Track length/Robot path (ft)	Throughput rate (/min)
Trad Robot	3.2	40
1 Bot	5.5	30
2 Bots	5.5	60
Trad Robot	3.8	35
2 Bots	10.5	37
3 Bots	10.5	55
2 waves of 2 Bots	10.5	74
Trad Robot	3.8	35
2 waves of 3 Bots	10.5	110
Trad Robot	3.8	35
2 waves of 4 Bots	11.5	137

Standard Robot

A standard robot must be programmed to make a somewhat similar motion in order to make a fair comparison (Fig. 11.55). This motion is the same whether a XYZ Cartesian, SCARA, or pedestal style robot. The robot travels down to the infeed conveyor, grabs a part, and moves to a location represented by the average dropoff location of the Trackbot. For comparing to the single bot above, the robot will have covered 18 in. replicating the No-Bot zone, leaving 22 in. to be covered at 4 ft/sec. The robot also must decelerate and reverse direction, approximately 0.5 sec. The 22in. will increase to 30 in. when comparing for waves of multiple Bots. Given that:

- 4 ft/sec max velocity;
- up/down motion 1 sec as per No-Bot zone;
- 22 in. remaining horizontal motion (0.5 sec).

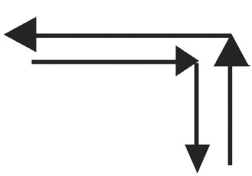


FIG. 11.55 Traditional robot's similar motion

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Two Bots

If one places two Bots on the same circular track, one Bot is in the No-Bot zone while the second Bot covers the four linear feet. The total distance has been chosen such that entering the No-Bot zone is not relevant. Given:

- 4 ft/sec speed chain;
- No-Bot zone of 1.5 ft;
- No-Bot segment takes 1 sec.

Two Bots on Oval Track

The next logical embodiment of track would be an oval. Two circular cogs on each end tension the chain. A standard 2 ft diameter wheel size is used. And since the two wheels cannot interfere with each other, the straight section of chain must be 2.2 ft long. This produces a total circumference of 10.5 ft.

Given:

- 4 ft/sec speed chain;
- No-Bot zone of 1.5 ft
- No-Bot segment takes 1 sec.

This produces a less than optimal arrangement. The extra distance takes more time and thus overall efficiency is reduced (Table 11.3). With only two Bots, the "next" one is nominally never close enough to the first one to gain any cycle time.

Three Bots on Oval Track

By adding a third Bot to the exact same track and conditions, almost the same level of performance of two Bots on a circular track can be duplicated. At first inspection, the additional size and Bot seems a waste, but the distance covered by the part has been increased. Table 11.3 compares a traditional robot's minimum distance. But the dropoff point can be much further away from the pickup point with the Trackbot with no additional cycle time penalty. A standard robot trajectory cannot be extended without cycle time penalty.

Two Waves of Two Bots

If the same oval track is used, a pair of two Bot waves can be placed effectively. The No-Bot zone becomes 1 ft longer because of the Bot pair.

Given:

- 4 ft/sec speed chain;
- No-Bot zone of 2.5 ft;
- No-Bot segment takes 1.25 sec.

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Two Waves of Three Bots

For two waves of three Bots, the same track length can be used. The No-Bot zone has increased to 3.5 ft, and the duration is now 1.5 sec, but the remaining 7 ft of track will be covered in 1.75 sec, so the system is compatible.

Two Waves of Four Bots

Two pairs of four Bot waves can now be investigated. Since the No-Bot zone is now 4.5 ft taking 1.75 sec, the remaining track length is only 6 ft, and at 4 ft/sec and 1.5 sec, the 10.5 ft track length is not compatible. The track would need to be increased to 11.5 ft overall.

The use of waves as the most efficient method becomes clear when one looks at the practical size of the No-Bot zone and the density possible if the wave of Bots loosely act as a unit. This is similar to congested highway driving, where if drivers are alert enough to stay on the tail of the car ahead of them during stop and go driving, the maximum throughput can be achieved.

The closed-loop track-based multihead robot discussed in this paper can have a significant impact on overall cycle time. For short pickup/dropoff distances, the system can be more efficient than a traditional robot. For longer distances such as 11.5 ft work and return total path length, the system can reach a factor of four times the throughput.

11.3.10. Distributed Control

First Control Architecture

Control and processing for the Trackbot system is distributed among multiple computers and processing units (Derby and Brown, 2002). This distribution of control allows for the quick response times necessary for real-time control and allows for simpler programming of the individual units. It also reduces the processing performance needs of any one individual unit. The Trackbot uses the following control and processing units:

- Host computer;
- Remote computer;
- Bots;
- Video acquisition and processing unit.

Overall control of the system is performed by the host computer, a standard PC running Windows NT (Fig. 11.56). On initialization the host establishes TCP/IP connections to both the remote computer and the video processing unit using standard Ethernet connections. After establishing the connections the host passes initialization and parameter information to each unit. By doing this, the host retains complete control and oversight of the entire system. In addition,

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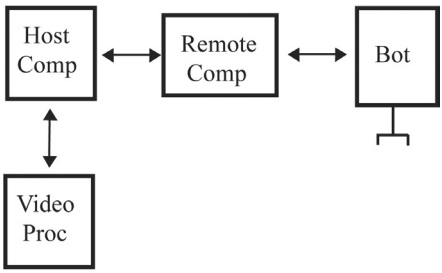


FIG. 11.56 Overall control block diagram

the host establishes a second TCP/IP connection with the remote unit to allow the remote unit to communicate urgent information to the host at any time. The host also has control of the various mechanical parts of the system, including the chain drive, air supply, and conveyors, through an I/O board. If an emergency condition is detected, the host can immediately stop the Bots.

When a user starts the processing operation the host starts a new control and processing thread. This thread determines where a wave of Bots should be sent to pick up product from the input conveyor and passes that information to the remote computer. Position information is sent for a complete wave of Bots each time, so that minimum Bot spacings are maintained to avoid collisions. The host is continuously updating the position information to the remote so that when a wave of Bots is ready to receive new commands, the most recent information is available. Attempting to have individual Bots plan their own buffering or obstacle avoidance is not possible with this architecture.

The remote computer is responsible for control of and communication with the individual Bots. The remote computer receives Bot parameter and product position information from the host over the TCP/IP connection established by the host. It then passes that information to individual Bots as they pass by the IR communication link. The remote computer is also responsible for maintaining spacing between individual Bots and waves of Bots. This is done by calculating and passing to the Bots additional delay times for the Bots when they stop to pick up product. The remote computer has approximately 125 msec to communicate with an individual Bot. This assumes a 4 ft/sec chain speed and approximately a 1 ft window in which the IR link between the remote computer and an individual Bot is allowed. During this time the remote computer must:

- Determine that a Bot has entered the IR window;
- Establish communication with the Bot;
- Confirm expected Bot identification;
- Calculate Bot spacing and delay, if any;

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- Request status information from the Bot;
- Pass any command, control, and parameter information to the Bot.

The remote computer detects that a Bot has entered the IR window when a flag on the Bot breaks a through beam sensor. A simple handshake and communication protocol is employed for the IR link, which includes a checksum for each message passed (Fig. 11.57).

Each Bot has a single board Pentium computer, flash memory, and an IR communication device located on top. Although a dedicated Pentium computer might seem overkill, the relatively cheap cost of the single board system, plus the ability to add a local vision system and have it processed efficiently, and the ability to add other functions, makes it more open for further expansions than a simple Basic Stamp processor or the like.

Each Bot operates independently from every other Bot, receiving its commands via serial IR from the remote computer. The Bot executes as a state machine as shown in Table 11.4. After initialization the Bot enters a state where it is waiting to establish an IR link with the remote computer. When this link is established, the Bot can receive requests for status and can receive updated parameters and command information. A command to pick up a product consists of a position and a delay time. After receiving a command the Bot looks for a flag that indicates the home, or zero, position. A position encoder on the

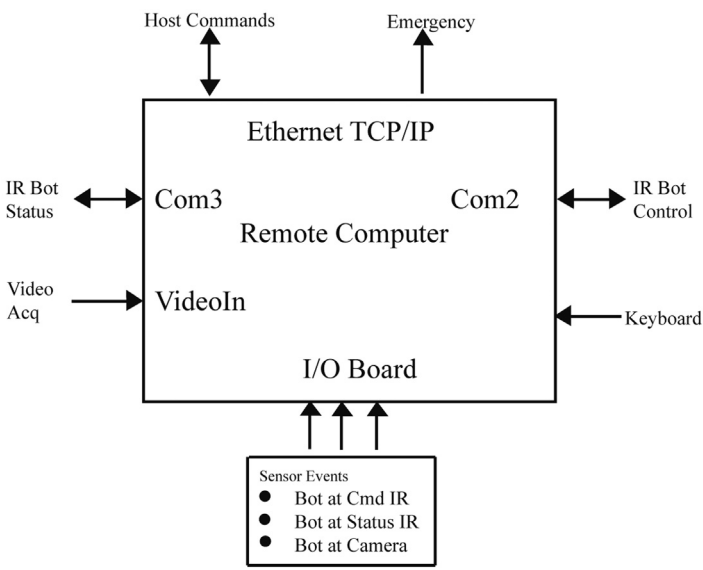


FIG. 11.57 Remote unit functionality

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TABLE 11.4 Bot Processor Functions

Thread	Description	Waits on	Signals
Main	State machine to perform Bot actions, dynamically update Bot parameters, communicate with remote via IR	IR input Sensor events Decoder event	
ISR	Sensor input interrupt routine	Sensor inputs Decoder counter	Sensor events Decoder event

moving cog is used to determine track position and in turn when to apply braking. The actual position at which the Bot finally stops is compared to the desired position in order to perform adaptive braking. This allows for variations in clutch slippage among the individual Bots to be accounted for and for accurate stopping positions to be possible. In a similar way, acceleration parameters are calculated when the Bot starts moving again. This allows for accurate times between when a Bot first applies its brake and when it finally achieves full speed again after stopping. This is important for maintaining Bot spacing and to avoid collisions between Bots. After the Bot has picked up a product it then drops it at the location it had previously received from the remote unit. The Bot processing then returns to the original state of waiting for an IR link. In a future system the Bot may also have a video input. This would allow the Bot to perform more precise positioning as well as rotating a product into a desired orientation before dropping the product off.

A vision system is used to determine the location of product on the input conveyor. An intelligent video acquisition and processing unit is used. This unit allows custom processing algorithms to be loaded and executed. The video unit continuously acquires new images and performs the processing to determine product location. The most recent locations are passed to the host computer on request, where they are subject to additional restriction criteria based on Bot spacing requirements. By determining product location in the video processor, the host computer only needs to analyze a small number of points (usually less than 10) as it requires them, as opposed to needing to continuously process 640×480 image points. The exact algorithm performed by the video unit can be modified and loaded onto the unit depending on the product being handled by the Trackbot.

11.3.11. Lessons Learned

The initial control scheme had some major difficulties. These needed attention before product commercialization. Most of the control problems were around

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TABLE 11.5 Host Processor Functions

Thread	Description	Waits on	Signals
Main	User interface	User	
Processing	Determines pickup position, transfers pickup request to remote, starts conveyor when required	Sleeps between remote queries	Conveyor timer
Emergency	Receives status from remote in event of emergency situation detected	Blocking read of emergency port socket	User (Stop chain?)
Conveyor timer callback	Check conveyor, stop it if full, reset timer if not	Timer	Conveyor relay

TABLE 11.6 Remote Unit Functions

Thread	Description	Waits on	Signals
Main (video)	Video processing, keyboard processing	BotAtCamera event video frame done keyboard input	Terminate event
Remote command	Receives commands from host	Blocking read of remote port socket Terminate event	IRCommand event Adds commands to Bot Command Block
IR command processing	Process commands and requests requiring communicating with Bots via IR. This includes commands in Bot Command Block, enumeration and updates to Bot parameters.	BotAtIR event IRCommand event Terminate event IR response Terminate event	
IR comm	Performs low-level IR communication	IR port input	IR response
Emergency	Sends emergency status to Host when signaled	Emergency event Terminate event	
ISR	Sensor input interrupt routine	Sensor inputs	BotAtIR event BotAtCamera event

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the restricted communication between the remote computer and the Bots and the issues of Real Time Control.

The IR communication "window" between the remote computer and a Bot was short compared to one lap around the track. The use of the separate remote computer to communicate with the Bots was chosen in order to minimize the real-time response time when a Bot passed the IR window and maximize the possible communication time. Even with this, the amount of information that could be passed between each Bot and the remote computer during that "window" was limited. As development continued it was found that additional status information was required from each Bot as it passed by. This information needed to be processed by the remote CPU and commands sent back to the Bot based on it. This included the need to anticipate what each Bot was going to do and to adapt based on how the Bot actually responded. There became a significant amount of information to transfer and confirm in a very small amount of time.

The brake control needed to handle the slipping of the drive clutch. As the hardware was broken in, the braking distance and acceleration characteristics changed. This, in turn, changed the positioning restrictions that needed to be accounted for by the remote CPU. The success or failure of a Bot in picking up product could also indicate problems with a particular Bot that would need to be handled as well.

The variable speed drive on the track chain also created some electromagnetic interference issues with the remote CPU and the IR serial communication device. This would sometimes require that information be re-sent, shortening an already short communication window. The IR device and driver circuitry was not meant to see industrial environments.

Second Control Architecture

Figure 11.58 shows the revised Trackbot Control Architecture. This includes what is really a simplification of all of the tasks and threads in the first architecture. By the improvement of RF wireless Ethernet technology to commercial readiness, the host CPU can talk to any Bot at any time. This eliminates the need for a real-time response when a Bot reaches a particular position (i.e., the IR communication window) and removes the limitation of a fixed amount of time to transfer information to and from a Bot. By doing this the remote CPU can be eliminated. The host computer now has complete control of the system, including the actual communication with the Bots. The Bots can also send status information back to the host (e.g., successfully picking up product), to which the host can respond (e.g., telling the Bot where to drop the product). The ability to communicate with any Bot at any time also allows the possibility of a more interactive command and control architecture between the host and the Bots in the future.

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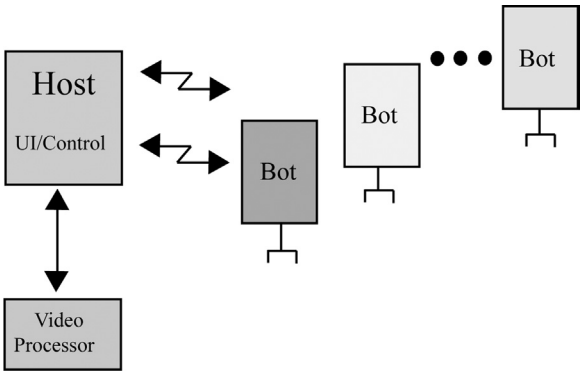


FIG. 11.58 Second control architecture

The Bots were improved with onboard backup batteries to eliminate some issues due to noisy pickups on the rails for the 12 V power supply. Each Bot's single board processor has an interface for an onboard vision camera, which can be added when needed. This allows for the picked up product orientation to be determined. The *Z*-axis has a rotary motion added to rotate the product for correct placement. The brake control was modified to remember the braking parameters from earlier runs, rather than determining them from initial conditions each time the Trackbot is turned on.

11.3.12. Trackbot Production and Market Acceptance

In early 2000, the author only had the initial testing model built. His company reached a development agreement with Cambridge Valley Machining Inc. (CVMI) of Cambridge, New York, in May of that year. After a very positive marketing visit to a major food producer, a major effort was undertaken to develop the prototype Trackbot in six months so as to make the November 2000 Pack Expo trade show. This is the premiere packaging trade show in the world, and it is held every two years.

So, the author, Dr. John McFadden (CEO of the author's company, Distributed Robotics LLC-DRLLC), and Mr. David Brown (controls and programming consultant) met weekly in design reviews with the CVMI team. This team included Mr. Donald Schneider, Sr. CVMI president, Mr. Merritt Bell, chief designer, and Mr. Jesse Ruppel, engineering intern. The entire group was starting with a clean sheet of paper, since nothing like the Trackbot had ever been built.

The fact that the prototype Trackbot, running at 1 ft/min rather than the target 4 ft/min, did get to Pack Expo was an engineering miracle. At the trade

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show, the marketing and sales company, American Productivity Group LLC (APGLLC), with Mr. Dwight Carey as principle, handled the contacts. APGLLC had many automation representatives attending the show, and 18–20 of them learned about the Trackbot.

The reactions at the show varied greatly. Some attendees loved the new technology, while some could never seem to understand it. Others wanted it to go much faster, and some were insistent on the use of vision (only the multiple line of sensors was implemented at this time). Still others needed a *Z*-axis rotation for their application. Some wanted to singulate empty Stand Up Pouches (SUP) to send them to pouch feeding machines, but no-one wanted to place down any amount of dollars to buy the first one. Since it was not a proven technology, no-one wanted to be the first. They all wanted to talk after the company had one in production in someone else's factory. This was a little surprising, since some customers want the first of any new product, but automation and packaging people are generally more conservative!

So the vision system described earlier was implemented, and the author was a speaker at the Pouches 2002 conference in the middle of that year. He visited several pouch equipment producers to gain information for a meaningful presentation on how the Trackbot fitted into the entire pouch material handing market. This conference focused on the SUP market, and was attended by over 300 of the industry users and suppliers. However, mid-2002 was already a slow market; a major machine producer had only sold two units in six months!

So the Trackbot production plans are complete, but the first production unit has not been built. Current DRLLC investors are looking towards the commitment of a Beta site customer, and the improvement in the economy for automation and capital equipment in general before proceeding.

11.4. OVERALL EXPERIENCES

The author's overall experiences of creating a startup company for the development and production of automation have seen ups and downs. Many times it has been exciting, when sharing a new method to solve an old problem; other times when the market goes flat, disappointing!

Here are some observations, many of which will echo generally held opinions, but when one encounters them for oneself, it does drive the message home:

1. Things take longer than expected. The development of the Stackbot at MBI and at CVMI always took longer than one had hoped. Some of this was due to components having 12–14 week lead times, some to the fact that each company was producing other products that had paying customers eagerly awaiting.

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2. It takes more dollars than expected. Both Stackbot and Trackbot took more money than initially planned. Increased feature desires from customers as well as early engineering attempts that were not satisfactory were responsible.
3. Business plans are easy to write, but hard to stick to. When in a business plan one anticipates a certain burn rate of dollars, and milestones that seem to slip away too easily, rewriting the business plan to reflect current reality too many times can discourage potential investors.
4. The market is fickle. Case studies of past development projects are good to learn from, but the market conditions can change in a very short time. The United States entry into Afghanistan in October 2001 helped to discourage one potential Stackbot customer, who the previous week wanted 14 units!
5. Which came first, the chicken or the egg? This conundrum lives on for potential investors wanting a startup company to get a few machines in production to feel good about investing. Yet a startup company needs funds to develop and produce the machines to be out there in the first place. DRLLC needs additional capital to make the next steps happen, but the 2002–2003 market has squashed the investment process.
6. Patents last for 20 years. This is a good thing when the market for automation (as well as many other markets) is so cyclical. The author's inventions are protected for many years until things are better, but the high costs of obtaining all of the international patents that might be lucrative have prevented DRLLC from the worldwide protection originally desired. Major automation companies like Adept Technologies, who had $100 million in sales of their robots in 2000, were forced to reduce their head count in early 2003 to around 25% of their peak when sales waned. Patents are a great reserve to draw on when rebuilding.
7. Inventing happens when it happens. One can brainstorm and work hard every day on a problem, and with more design and build experiences, coupled with learning from others' work, one can see the reapplication of some traditional concepts into new areas. However, some divine inspiration does help now and again. That proverbial light bulb going off over one's head, or the little voice from within giving one a vision on how to proceed is priceless.
8. Licensing technology is difficult. A common strategy for inventors is to license the invention to some large corporation and collect royalties. The author was initially successful with the Stackbot at MBI, but that did not last. Licensing the Dockbot has proven to be frustrating.

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9. Never say never. After GRASP closed its doors, the author told everyone who would listen that he would never start another automation company, but when the new inventions (Dockbot, Stackbot, and Trackbot) became clear to him, he felt a strong sense to go ahead and do it again. This included raising capital, a fate he and most engineers dread!
10. Do not ever give up.

Conclusions. Outside, uncontrollable factors can have a profound effect on the automation designer.

QUESTIONS

1. Based on the technology of the Dockbot, Stackbot, and Trackbot, what range of devices could be made from the combination of two or three of these? What would their markets be?
2. Based on the mix of technology of the Dockbot, Stackbot, and Trackbot in 1998, what could an alternative approach to machine development and bringing them to market be, if one knows that there will be a giant economic slowdown in 2002–2003?
3. What operations could the Trackbot achieve in the material handling sortation business, as in USPS, UPS, and FEDEX?
4. What are the limitations of the Stackbot's track plate such that it is difficult to handle more than two pallets simultaneously?
5. What is the lower limit of number of Bots one could use on a Trackbot and why? Why would this be practical compared to a standard robot?
6. How would a Dockbot handle a smaller Dockbot for nano-assembly?
7. Sketch how a Dockbot would operate in conjunction with a *XY* table.
8. Sketch what a two-arm robot version of a Dockbot might look like.
9. Design an alternative method of driving a Trackbot Bot clutch pack.
10. What could be done businesswise in 2003 for Distributed Robotics LLC to improve the situation?

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